The trading-signal engine

A technical audit — what it computes, what it sends, what was decided, and what is wrong with it

Subject Self-hosted Ghostfolio fork · trading-signals engine · `apps/api/src/services/signals`

Date 27 August 2026 · **Code state** working tree at `e7fa40ed5` plus uncommitted work through 2026-08-26

Evidence 200 real `SignalLog` rows · 1,306,576 `MarketData` rows · 1,075,012 `OhlcBar` rows · 936 symbol profiles · 787 MB database

Method Every figure is either read from the live database or recomputed by loading the engine's own compiled services and re-running them. Nothing is quoted from memory.

Three defects were found, all live. They are stated in §1.3 and developed in §10.

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1 · Scope, method, and the headline findings

1.1 What was audited

This is an audit of a live system, not a description of one. The distinction matters: every formula below was re-executed against the engine's own compiled code and the numbers it actually wrote, and the report records where the two disagree.

In scope: the signal engine under `apps/api/src/services/signals` (34 files, ~18,600 lines), the constants in `libs/common/src/lib/config.ts` that parameterise it, the commission model in `libs/common/src/lib/nordnet-fees.ts`, the two price stores, the scheduling and queue layer, and the Telegram delivery path.

Out of scope: the upstream Ghostfolio portfolio, accounting and import code the fork inherits; the Angular client beyond the surfaces that render signal output; the Telegram conversational assistant (§10 of the operations guide); and the fund/NAV sleeve, which is not signal-driven.

1.2 How to check any number here

Three scripts produced everything in this document. All are read-only.

Script	What it does
`extract-evidence.cjs`	Pulls the exemplar `SignalLog` rows, then re-runs `IndicatorsService` and `ForecastService` — the compiled artefacts, not a reimplementation — over a point-in-time reconstruction of the price series, and diffs every field.
`measure-duplicates.cjs`	Computes each symbol's indicator snapshot twice: once over the series as the engine reads it, once over the same series collapsed to one row per calendar day.
`build-audit-pdf.cjs`	Typesets this document. KaTeX runs at build time; nothing is fetched at print time.

The reconstruction is genuinely point-in-time. It bounds the window by `SIGNAL_HISTORY_FETCH_DAYS` exactly as the engine does, and filters on `MarketData.createdAt` so rows written *after* a signal fired cannot leak into its reproduction. That second constraint turned out to be load-bearing, for a reason that is itself one of the findings.

1.3 Findings, up front

The engine's mathematics is sound and reproduces exactly. §7.1 recomputes a real DIP signal and matches all eleven stored fields to five decimal places. The defects are not in the formulas; they are in the series the formulas are fed.

FINDING 1 — LIVE, LARGEST IMPACT

Two price rows exist for the same symbol and calendar day, across 527 symbols and roughly 30 consecutive sessions. The universe importer stamped its rows at the exchange session time (13:30 UTC for New York, 07:00/08:00 for Europe); the hourly gatherer stamps UTC midnight. `@@unique([dataSource, date, symbol])` is on a `DateTime`, so the two never collide. Every indicator window counts array positions, so over the affected range a "50-day average" spans about 25 sessions.

Measured over 397 affected symbols: **the composite score changes for 364 of them** when duplicates are removed (mean absolute change 5.6 points, maximum 24); **the MACD histogram changes sign for 98**; the 30-day high — which sets the dip trigger — **is understated for 137, by up to 35%**. §10.1.

FINDING 2 — LIVE

Every intra-session evaluation counts the current day twice. The gatherer writes a row for today as soon as it runs, initially carrying the previous close. That row is a weekday, so it survives the trading-day filter, and the series ends with the previous close repeated. This is not a hypothesis: it is how §7.1 achieves an exact reconciliation, and without modelling it the reproduction fails. Wilder's RSI is provably invariant to the extra bar; Bollinger %B and MACD are not. On the audited signal it moved %B by 8.4%, the MACD histogram by 25.7%, and the composite score from 76 to 75. §10.2.

FINDING 3 — LIVE

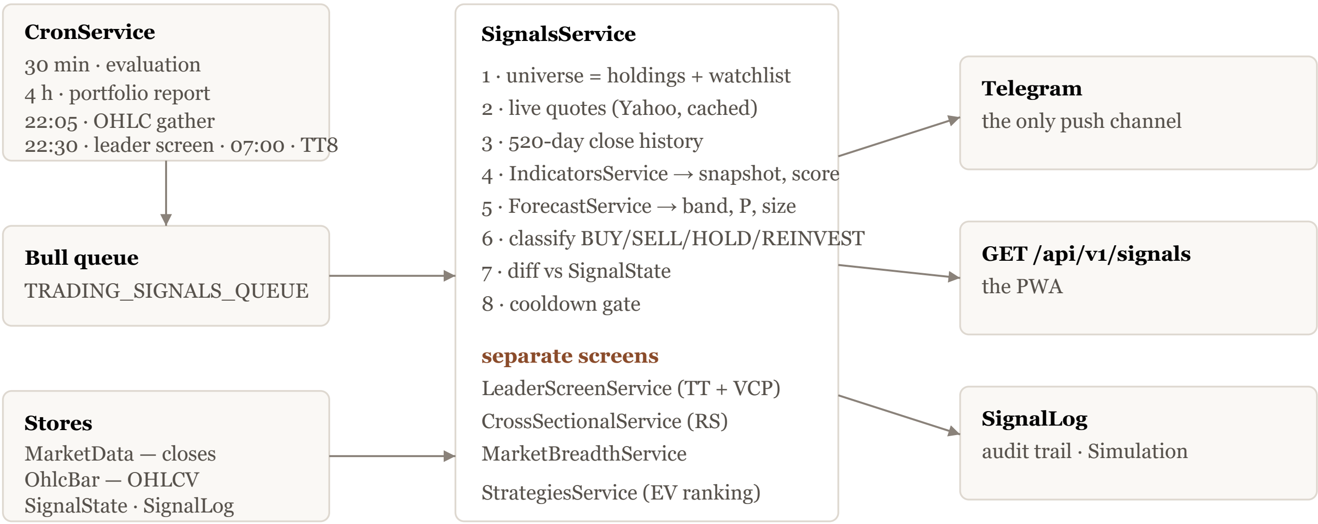
The suggested position size is computed in USD and displayed beside a native-currency price, with no currency label. On 2026-08-24 the engine sent "Suggested: 66.85" for a Danish stock quoted at 462.50 DKK, and the same 66.85 the following day for a Swedish stock at 169.50 SEK and a Swiss stock at 726 CHF. The intended amount is 66.85 *US dollars*. Read as SEK, that is an instruction to buy roughly one-ninth of the intended position. §10.3.

Three further observations are recorded rather than raised as defects: the logged take-profit on a BUY row is not fee-adjusted while the exit machine's is (§10.4); a class-level doc comment still calls the leader screen "the engine's PRIMARY buy path", which the engine's own evidence overturned three days later (§10.5); and `SIGNAL_TREND_TEMPLATE_PREFERRED_RS` remains defined but is no longer read on the path it was written for (§10.6).

2 · The engine as it runs today

2.1 Shape

A scheduler enqueues jobs; a Bull processor drains the queue and resolves each non-demo user; `SignalsService` orchestrates everything else. There is exactly one user.



The runtime. Everything left of `SignalsService` is infrastructure the fork inherited; everything inside and right of it is the fork's own.

2.2 Cadence

Each surface fires at a rate matched to how often the thing underneath it changes. That is a design decision, not an accident: the leader alert exists in its present form only because an earlier version re-sent the same two dozen names every evening.

Scheduled work, from `cron.service.ts`. Times are the host's local zone (CET).

When	Cron	Job	What it sends
every 30 min, weekdays	`*/30 * * * *`	evaluation	DIP / REVERSAL / SELL, on category change only
every 5 min, weekdays	`*/5 * * * *`	intraday trailing check	exits for tracked positions past their frozen target
every 4 h, weekdays	`0 */4 * * *`	portfolio report	heartbeat — always sends
22:05, weekdays	`5 22 * * 1-5`	OHLC refresh	nothing; it writes bars
22:30 + 10 min delay	`30 22 * * 1-5`	leader screen	VCP breakouts only, 5-day per-symbol cooldown
07:00, weekdays	`0 7 * * 1-5`	Trend Template entrants	new 8/8 crossings at RS ≥ 90, 10-day cooldown
1st and 15th, 07:00	`0 7 1,15 * *`	shortlist digest	the standing 8/8 ∩ RS ≥ 90 list, capped at 12
Mondays 12:00	`0 12 * * 1`	fund recommendation	which index fund to add
25th, 09:00	`0 9 25 * *`	monthly plan	the 750 USD deployment split

The 22:05 → 22:30 ordering is deliberate and tight. The gather must finish before the screen reads the bars it writes, and 22:05 is late enough that both the European (~17:30) and US (22:00) closes have settled.

2.3 Why the boot catch-up exists

A `@nestjs/schedule` cron does not catch up. A firing missed because the host was asleep is lost, not deferred — and the host is a laptop that is routinely asleep at 22:05. For an alert that is merely a missed notification. For the gather it is a permanent hole in the bar history that nothing else fills, which is why `runBootCatchUps` runs OHLC refresh → leader screen → shortlist → entrants *in sequence* on startup, each isolated so one failure cannot cancel the rest.

An earlier version staggered these on 30-second timers and hoped the gather would win the race. Over ~900 symbols the gather takes minutes, so it lost. Sequencing replaced the stagger.

3 · The data layer

3.1 Two stores, and why

Measured 27 August 2026.

Store	Holds	Rows	Symbols	Newest	Written by
`MarketData`	one close per *calendar* day, weekends forward-filled	1,306,576	949	2026-08-27	hourly `gather7Days()`
`OhlcBar`	open/high/low/close/volume per session	1,075,012	864	2026-08-27	22:05 `refreshOhlcBars`

The split is not redundancy. `MarketData` is upstream Ghostfolio's and carries closes only; RSI, MACD, Bollinger, the moving averages and close-to-close volatility all run off it. Everything that needs a high, a low or a volume — ATR, the Trend Template's 52-week extremes, the VCP's swing detection and both of its volume tests, Yang-Zhang volatility — reads `OhlcBar`, which the fork added.

Until 2026-08-25 `OhlcBar` had *no writer inside the running application*. `OhlcBarService.upsertMany` had zero callers in `apps/`; the bars advanced only when someone remembered to run a backfill script. Because `MarketData` stayed current the whole time, prices moved, the composite score moved, the sparklines moved — and the Trend Template, the VCP detector and the RS percentile read a four-day-old market while looking perfectly alive. The nightly gather (§2.2) exists to close that.

3.2 The trading-day problem, and its sequel

`MarketData` stores one row per calendar day: the gatherer walks forward a day at a time with no weekday check and carries the last known price into Saturday and Sunday, writing them `state: 'CLOSE'` — indistinguishable from a real session. Every window in `IndicatorsService` is a bare array-index count whose *name* means trading days. On a raw calendar series each one silently spans about five-sevenths of the sessions it claims, so a 200-day average becomes a 138-session average, and daily volatility is deflated by $\sqrt{5/7} \approx 0.845$.

`toTradingDayCloses` fixes that by filtering on `getUTCDay()` before the closes reach any indicator. Filtering by weekday rather than by price equality is deliberate: it is deterministic, symbol-independent, and cannot discard a genuine flat close. About nine market holidays a year survive as zero-return weekday rows, leaving a residual understatement near 1.7% instead of 17%.

> **THE SEQUEL — FINDING 1**
>
> That defence assumes one row per weekday. It no longer holds. The universe importer wrote `date: new Date(timestamps[i] * 1000)` straight from Yahoo's chart response — the session-open epoch, 13:30 UTC for New York — while the gatherer writes UTC midnight. Both are weekdays, both survive the filter, and the unique constraint sees two distinct `DateTime` values.
>
> The result is arithmetically the same failure `toTradingDayCloses` was written to prevent, arriving through a different door, and worse over the affected range: not five-sevenths of the sessions, but one half. Full measurement in §10.1.

3.3 The universe

936 symbol profiles. Membership is *generated*, not hand-maintained: `index-constituents.ts` is produced by a script and committed as source, so the build never depends on those URLs being reachable.

Cohort	Names	Source
S&P 500	503	generated, mechanical
EURO STOXX 50	50	generated, mechanical
European mid-caps	100	**curated** — FTSE 100/250 + STOXX Europe 600, filtered on cap, liquidity, forward P/E
Personal watchlist, ETFs, funds	~283	manual

Universe size is a direct input to signal quality now that the primary cross-sectional metric is a percentile: a rank over 334 names is a blunter instrument than one over 750. The mid-cap block also closes a real gap — the EURO STOXX 50 is euro-area only, so before it the database held three UK listings in total.

The curation has a cost, recorded rather than hidden: selecting universe members on fundamentals means any *backtest* over those names inherits the selection. Forward signals on them are clean; the comparable historical subset stays the mechanical S&P / EURO STOXX one.

3.4 Currency

Each stock's BUY/SELL logic runs in its **native** currency — an SEK stock is judged in SEK. USD is the base currency for cross-asset ranking and portfolio totals. This is the right split, and it is where Finding 3 lives: one quantity, the suggested position size, crosses the boundary without converting or labelling.

4 · The metric dictionary

Every number the engine computes, what it means, where it is calculated, what threshold it faces, and how it fails. Notation throughout: P_t is the close on trading day t, with P_n the most recent; $H = 42$ is the horizon in trading days; σ is daily volatility.

4.1 Tier 1 — the technical indicators

All of these are implemented directly in `IndicatorsService` — no external library, no Python, no model. They take a plain array of closes, which is all `MarketData` stores.

Relative Strength Index, 14 periods (Wilder)

Gains and losses are separated, seeded with a simple mean over the first 14 changes, then smoothed recursively:

$$\overline{G}_t = \frac{13\,\overline{G}_{t-1} + G_t}{14}, \qquad \overline{L}_t = \frac{13\,\overline{L}_{t-1} + L_t}{14}, \qquad \mathrm{RSI} = 100 - \frac{100}{1 + \overline{G}_n/\overline{L}_n}$$

Because the recursion runs from the first bar of the supplied array to the last, RSI in principle depends on how much history is passed. In practice, with ~350 bars, the seed's influence has decayed to nothing: §7.1 reproduces a stored RSI to seven significant figures from an independently reconstructed series.

One property of this recursion is worth stating because it becomes diagnostic in §7.1. Appending a bar with *zero change* multiplies both averages by 13/14, leaving their ratio — and therefore the RSI — **exactly unchanged**. Wilder's RSI cannot see a duplicated close. Bollinger %B and MACD can.

Read as: low is oversold, and in this engine oversold is *attractive*. The composite score weights (100 − RSI) at 0.25.

MACD, 12/26/9

$$\mathrm{MACD}_t = \mathrm{EMA}_{12}(P)_t - \mathrm{EMA}_{26}(P)_t, \qquad \text{signal} = \mathrm{EMA}_9(\mathrm{MACD}), \qquad \text{hist} = \mathrm{MACD} - \text{signal}$$

The EMAs are seeded at P_0 rather than with a simple mean over the first period, so like RSI they are formally history-dependent and practically converged. The histogram enters the score as a *binary*: 100 if ≥ 0, else 30. That step function is why a sign flip is worth 15 × 70 / 100 = 10.5 points of composite score, and why Finding 1's 98 sign flips matter as much as they do.

Bollinger %B, 20 periods, 2σ

$$\mu = \frac{1}{20}\sum_{i=n-19}^{n} P_i, \quad s = \sqrt{\frac{1}{20}\sum_{i=n-19}^{n}(P_i - \mu)^2}, \quad \%B = \frac{P_n - (\mu - 2s)}{4s}$$

Note the population divisor $1/20$, not the sample $1/19$. That is a convention choice, consistent with most charting packages, and it is applied consistently. %B is 0 at the lower band and 1 at the upper; it is *not* clamped inside `bollinger()`, but the score clamps it to $[0, 1]$ before use.

Moving averages and momentum

SMA_{50}, SMA_{150}, SMA_{200} are plain means of the trailing window. Momentum is a simple percentage change over 63 and 252 array positions. `smaSeries` computes the rolling average incrementally so its slope can be measured, which criterion 3 of the Trend Template needs.

`slopeUpDuration` counts consecutive non-decreasing bars back from the end. This answers a *duration* question rather than a "higher than N bars ago" one — a single sharp uptick after a long slide passes the latter and fails this. That distinction is the whole point of the criterion.

Volatility — three estimators

The close-to-close estimator is the default and the one that sets every price band:

$$r_t = \ln\frac{P_t}{P_{t-1}}, \qquad \sigma = \sqrt{\frac{1}{N-1}\sum_t (r_t - \bar{r})^2}$$

with N pinned to `SIGNAL_VOLATILITY_LOOKBACK_DAYS` = 252 rather than "however long the caller's array happens to be". That pinning is deliberate: σ sets every stop, target and trailing band, so letting it drift with an unrelated fetch-window constant would move those levels for reasons that have nothing to do with the market.

Two OHLC estimators exist for the cases where bars are available. **Garman-Klass** uses the intraday range and is roughly 7× more efficient than close-to-close, but is blind to overnight gaps. **Yang-Zhang** adds an overnight-return variance and a Rogers-Satchell term:

$$\sigma^2_{YZ} = \sigma^2_{\text{overnight}} + k\,\sigma^2_{\text{open-close}} + (1-k)\,\sigma^2_{RS}, \qquad k = \frac{0.34}{1.34 + \frac{n+1}{n-1}}$$

It is the most efficient estimator available from daily OHLC and, unlike the others, is unbiased in the presence of opening jumps. The engine uses it for exactly one purpose: the stop/target/trail bands of **owned active trades**, where a stable estimate matters most and the symbol count is small enough to afford the bar fetch. Everywhere else — including every entry signal — the close-to-close estimator runs. That asymmetry is defensible but undocumented, and it means a position's bands can shift the moment it is flagged as an active trade.

ATR (Wilder-smoothed)

$$\mathrm{TR}_t = \max\left(H_t - L_t,\ |H_t - P_{t-1}|,\ |L_t - P_{t-1}|\right)$$

seeded with the mean of the first 14 true ranges and then Wilder-smoothed, so the printed number matches what TradingView shows — a stop derived from a different ATR than the one on screen is a trap. ATR is computed and surfaced but does **not** currently set any live stop. §11.2 argues it should.

4.2 The composite score

A single 0–100 "buy attractiveness" number, built as a weighted blend so that several signals must agree before a candidate ranks highly:

$$S = \frac{1}{\sum w}\left[0.25(100 - \mathrm{RSI}) + 0.30\,(1 - \%B^{\,\mathrm{clamped}}) \cdot 100 + 0.15\,m + 0.15\,\tau + 0.15\,\mu\right]$$

Term	Weight	Value	Reads as
RSI	0.25	`100 − RSI`	oversold is attractive
Bollinger	0.30	`(1 − %B) × 100`	near the lower band is attractive
MACD m	0.15	`hist ≥ 0 ? 100 : 30`	momentum turning up
Trend τ	0.15	`P ≥ SMA200 ? 100 : 20`	healthy long-term uptrend
Momentum μ	0.15	`mom12M ≥ 0 ? 100 : 30`	quality filter

The normalisation by $\sum w$ (rather than by 1.0) means a symbol missing an input — no 200-day average yet, say — is scored on the terms it has rather than penalised for the absence. The result is rounded to an integer.

> **THE SCORE'S SIGN IS THE ENGINE'S OLDEST ARGUMENT WITH ITSELF**
>
> Two of the five terms reward *weakness*, and they carry 55% of the weight. This is not an oversight — it is what makes the score a dip-finder. The 2026-08-21 Minervini work was built on the hypothesis that this was the engine's core defect. The hypothesis was tested and rejected: the dip entry beats the base rate at every horizon (§9.2). The score's sign survived the challenge on evidence.

The score **ranks** candidates and, since 2026, also **gates** them: `SIGNAL_BUY_SCORE_MIN = 55`. A gate on a quantity whose mean absolute error is 5.6 points (Finding 1) is a gate that misfires in both directions.

4.3 The horizon band, and the four levels that hang off it

Every price level in the engine is expressed in one unit: the expected move over the trading horizon.

$$b = \sigma\sqrt{H}, \qquad H = \texttt{SIGNAL\_HORIZON\_DAYS} = 42$$

Forty-two trading days is about two months. The $\sqrt{H}$ scaling is the whole reason the targets are reachable. A stock with 1.5% daily volatility moves about 9.7% at one sigma over 42 days, so a flat +30% target is a three-sigma event that essentially never fires inside the window you actually trade in. Scaling to each name's own band keeps the target attainable *and* makes two different stocks' targets comparable.

The four levels. All four are multiples of the same b.

Level	Formula	Constant	Applies to
Buy (dip) level	$\min\big(H_{30}(1-d),\ H_{30}(1-k\,b)\big)$	$d = 10\%$, $k = 1.5$	watchlist candidates
Take-profit	$(\bar{P}+f)\big(1+\max(\phi,\ 1.5\,b)\big)$	$\phi = 13.4\%$	active trades
Stop-loss	$\bar{P}\,(1-2b)$	$2\times$	active trades
Trailing stop	$\text{peak}\,(1-1\,b)$	$1\times$	after the target is touched

where H_{30} is the 30-day high, $\bar{P}$ the average buy price and f the round-trip commission divided across the held shares, so the target clears the fee as well as the profit goal.

Three details are easy to miss and all three matter:

- **The buy level takes the *more conservative* of two references.** For a calm stock the 1.5σ leg is shallower than 10%, so the fixed floor binds; for a volatile one the sigma leg binds and demands a deeper drop. Calm names cannot trigger on noise.
- **ETFs are calibrated separately — $d = 5\%$, $k = 1.0$.** Before this, no ETF signal had *ever* fired: a diversified ETF essentially never prints a 10% drop off a 30-day high outside a crash (broad index ETFs sit 1–3% off), and the 1.5σ leg pushed volatile thematic ETFs to 20–40% required drops.
- **The stop is twice the band and the trail is one band.** The asymmetry is intentional: the stop must survive normal noise while the trail should exit promptly once a target has already been banked.

There is no day-count exit. A position can sit open past the 42-day horizon indefinitely if price never crosses a level — the horizon sizes the bands, it does not time the trade.

4.4 Tier 2 — forecast, probability, sizing

These are context, not predictions: a calibrated band and a probability, never a point price target.

EWMA volatility (RiskMetrics)

$$\sigma_t^2 = \lambda\sigma_{t-1}^2 + (1-\lambda)r_t^2, \qquad \lambda = 0.94$$

seeded at r_0^2. The effective memory is about 16 days, so this reacts far faster than the 252-day sample σ. The engine deliberately uses *different* volatilities in different places: the sample σ sets the price bands (stability), the EWMA σ drives the probability and forecast band (responsiveness). For the audited DVA signal these differ by a factor of 1.27 — 43.8% annualised against 55.5% — precisely because the stock had just fallen 28%.

Drift, and why it is set to zero

`ForecastService.drift` exists and clamps the mean daily log return to ±0.2%/day. **Every caller passes 0 instead.** That is the engine's central honesty commitment, stated in the code as "the drift problem": a drift estimated from a short history is dominated by noise and biases probabilities toward whatever just ran up. Setting it to zero makes the probability risk-neutral and, more importantly, comparable across the whole universe.

The direct consequence is that expected value is *usually negative* and the system says "nothing compelling" far more often than it says "buy". That is the design working, not failing.

Expected-move band

$$\big[\,P\,e^{\mu h - k\sigma\sqrt{h}},\ \ P\,e^{\mu h + k\sigma\sqrt{h}}\,\big], \qquad \mu = 0,\ k = 1,\ h = \texttt{SIGNAL\_FORECAST\_HORIZON\_DAYS} = 252$$

Note $h = 252$, not 42. The forecast band and the reach probability run over the *expected holding period* — a year — while the price levels are sized on the 42-day band. Two different horizons live side by side in one signal, which is defensible (they answer different questions) but is not signposted anywhere in the delivered message.

Reach probability

The analytic terminal probability under a lognormal model, not the Monte Carlo touch probability:

$$z = \frac{\ln(T/P) - \mu h}{\sigma\sqrt{h}}, \qquad \Pr[\,P_h \geq T\,] = 1 - \Phi(z)$$

with Φ evaluated through an Abramowitz & Stegun 7.1.26 `erf` approximation (max error ≈ 1.5 × 10⁻⁷). A Monte Carlo *touch* probability is also implemented (`hitTargetProbability` , 2,000 GBM paths, Box-Muller normals) but is not used on the signal path — the analytic terminal figure is used everywhere so signals and strategies stay directly comparable, and because it is cheap enough to run for every symbol on every pass.

Terminal is a strictly harsher test than touch: a stock that spikes through the target and falls back counts as a miss. That is the conservative choice, consistent with drift = 0.

Position sizing

$$A = B \cdot \min\left(0.25,\ \frac{0.20}{\sigma_{\text{ann}}}\right), \qquad \sigma_{\text{ann}} = \sigma\sqrt{252}$$

A volatility target of 20% annualised with a hard 25% cap on the cash budget B. More volatile names get less capital.

In practice the cap binds almost always, because the volatility leg only bites above 80% annualised. Measured across all 107 sized DIP and REVERSAL signals in the log, **103 are exactly** $0.25B$ — the three distinct values 25.00, 66.85 and 207.50 correspond to cash balances of \$100, \$267.40 and \$830. Only four signals were ever sized by volatility rather than by the cap. The sizing rule is, in effect, "a quarter of the cash", and the elaborate part of the formula has fired four times.

B is `balanceInBaseCurrency` — US dollars. The signal it decorates is in the asset's native currency, and **85 of the 147 BUY signals in the log are not in USD**. That is Finding 3.

4.5 Expected value

$$\text{EV} = p \cdot g - (1 - p) \cdot s$$

with p the reach probability, $g = \max(13.4\%, 1.5b)$ the target gain and $s = 2b$ the stop loss. This is the ranking quantity — picks are ordered by EV, not by the composite score, which is only an eligibility gate.

The geometry is worth making explicit, because it explains why EV is nearly always negative. Since $g \approx 1.5b$ and $s = 2b$, the payoff ratio is fixed at about 3:4 regardless of the stock. Break-even therefore needs

$$p^* = \frac{s}{g + s} = \frac{2b}{3.5b} \approx 0.571$$

A 57% terminal probability of a 1.5σ move, with zero drift, is not something the lognormal model will often produce — under $\mu = 0$ the terminal probability of exceeding a $+1.5b$ move over the same horizon that defines b is well under a half by construction. **Negative EV is structural, not empirical.** The ranking is still meaningful — it orders candidates by how *close* to break-even they get — but the sign of the number carries no information, and reading "EV = −0.148" as a prediction of a 14.8% loss would be a serious misreading.

4.6 The Trend Template — eight criteria

Minervini's screen, implemented in `LeaderScreenService.trendTemplate`. It requires at least 221 bars (200 for the average plus 21 for its slope), returns `null` below that, and reports the raw number each criterion was decided on so a near-miss stays legible.

The eight criteria and the constants that hold their thresholds.

#	Criterion	Test	Constant
1	above all moving averages	$P > \mathrm{SMA}_{50} \land P > \mathrm{SMA}_{150} \land P > \mathrm{SMA}_{200}$	—
2	150 above 200	$\mathrm{SMA}_{150} > \mathrm{SMA}_{200}$	—
3	200-day rising	non-decreasing for ≥ 21 consecutive bars	`SMA200_RISING_DAYS`
4	stack in order	$\mathrm{SMA}_{50} > \mathrm{SMA}_{150} > \mathrm{SMA}_{200}$	—
5	above the 50-day	$P > \mathrm{SMA}_{50}$ — overlaps #1, both are in his list	—
6	off the 52-week low	$(P - L_{52})/L_{52} \geq 0.30$	`MIN_ABOVE_LOW_PCT`
7	near the 52-week high	$(H_{52} - P)/H_{52} \leq 0.25$	`MAX_BELOW_HIGH_PCT`
8	relative strength	$\mathrm{RS} \geq 70$ percentile	`MIN_RS`

"8/8" means all eight. Criterion 8 is the only cross-sectional one and cannot be derived from a single symbol's history; passing `rsRank = null` evaluates the seven price criteria and fails the eighth, which is the correct behaviour for a universe too small to rank. A cold rank cache therefore renders **unranked**, not *failed* — the same distinction the RS service already draws for a newly listed stock.

> **WHY CRITERION 5 MAKES DIP AND THE TREND TEMPLATE MUTUALLY EXCLUSIVE**
>
> A dip is defined as at least 10% (or 1.5σ) below the 30-day high, which puts price under the 50-day average essentially always. Measured over 632 real DIP days: criterion 5 passes on **0.3%** of them. The maximum observed pass count on a dip day is 6/8; 7 and 8 never occur.
>
> The two entries are not merely differently tuned — they are structurally disjoint. Gating one on the other, which was the plan on 2026-08-21, produces exactly zero signals. This is the single most important structural fact about the engine and it is the reason the two paths are separate surfaces rather than a stack.

4.7 The VCP detector

Minervini's Volatility Contraction Pattern is a pattern chartists read visually; what follows is a quantitative proxy, and it will disagree with the eye in both directions. The pipeline runs over the trailing 60 bars:

Merge-then-suffix. The middle two stages are the 2026-08-24 rewrite; what preceded them is §9.3.

Merging fuses adjacent pullbacks that a weak rally failed to separate. If the rally between two troughs recovers less than `SIGNAL_VCP_MERGE_RALLY_PCT` (50%) of the prior drop, the two become one contraction spanning the first peak to the *lower* of the two troughs. Merging cannot conceal a low, because the deeper trough becomes the merged depth.

The suffix rule then takes the maximal *contiguous* run at the end of the sequence whose depths shrink monotonically and whose lows rise. Contiguous is the load-bearing word: a lower low truncates the base rather than being stepped over.

The gates, in the order they are applied. Failing any one returns a structure with `isValid: false` *and a human-readable reason.*

Gate	Threshold	Rationale
contraction count	3 ≤ n ≤ 6	fewer is not a pattern; more is "more base than setup"
final tightness	≤ 10%	Minervini asks 3–6%; measured median is 3.5%
base length	20 ≤ days ≤ 60	shorter is "too young to be accumulation"
base depth	≤ 35%	deeper is "a broken stock, not a consolidation"
volume dry-up	≤ 0.85×	**hard gate** — above it there is no valid VCP at all

Two volume tests, not one

This is the subtlety most often collapsed, and the engine deliberately reports both.

$$\text{dryUpRatio} = \frac{\overline{V}_{\text{final contraction}}}{\overline{V}_{50}}, \qquad \text{breakoutVolumeRatio} = \frac{V_n}{\overline{V}_{50}}$$

Dry-up is a *hard gate* at 0.85 — any base the engine shows you has already passed it, and nothing in the Trend Template can compensate. Breakout volume is an *event* test at 1.40, applied together with $P_n >$ pivot.

The pivot is the high of the final, tightest contraction. Status is one-sided by design:

Status	Condition
`BREAKOUT`	$P_n >$ pivot **and** volume ≥ 1.40×
`FAILED_BREAKOUT`	$P_n >$ pivot **and** volume < 1.40× — supply absorbed the move
`AT_PIVOT`	$P_n \leq$ pivot and within 2% below it
`FORMING`	everything else

Making `AT_PIVOT` one-sided — still *below* and closing in — was a correction, not a preference. A symmetric ±2% test filed the textbook breakout failure as "at pivot": 45 of 110 at-pivot names, 41% of a bucket meant to hold names still approaching.

One reporting decision worth recording: `dryUpRatio` is surfaced only when the VCP is valid. The rejection path hard-codes `dryUpRatio: 0`, and a rendered 0 is indistinguishable from "extremely dry" when it actually means "never measured". Where dry-up *was* the rejection cause, the ratio appears in the rejection reason instead.

> **INTERPRETATION VS MEASUREMENT**
>
> The thresholds above, and Minervini's "40–50% above average" behind the 1.40, are what the engine computes. Reading volume as a proxy for institutional participation is the conventional interpretation of those numbers — reasonable and widely used — but this engine observes *how much* traded, never *who* traded. Nothing in the data distinguishes an index fund from a portfolio manager.

4.8 Relative strength — the only cross-sectional metric

RSI, MACD, Bollinger and the composite score all answer "how does this stock compare with its own past?". None can answer "how does it compare with every other stock right now?", which is what Minervini's criterion 8 and O'Neil's RS Rating both depend on.

$$\text{rsScore} = 0.4\,R_{3m} + 0.2\,R_{6m} + 0.2\,R_{9m} + 0.2\,R_{12m}$$

then converted to a 1–99 percentile across the universe. The double weight on the most recent quarter follows IBD's published approach: the rank responds to a stock turning up without being whipsawed by a single strong month. Periods are counted in trading days (63/126/189/252), not calendar days.

Two guards are worth naming. Names with under 253 bars are **excluded** rather than given a low rank — a newly listed stock is *unranked*, not *weak*, and conflating those would quietly bias the screen against recent IPOs. And a universe below `SIGNAL_RS_MIN_UNIVERSE` = 30 returns scores with no percentile at all, because ranking five names 1–99 is arithmetic theatre.

Look-ahead is the real risk here, and the code is built around it. A cross-sectional rank computed from the full series and then applied to a historical date silently encodes the future, and the resulting backtest is worthless in a way that looks perfectly healthy. Hence `asOf` on every entry point, a hard filter on `date <= cutoff`, and a regression test asserting that appending future bars does not change a past ranking. `momentum12m2` — the 12-month return excluding the most recent month — is computed and carried but does not enter `rsScore`; §11.4 argues it should.

`peerRankMap` additionally ranks each name inside its sector and each sector against the others, so a row can read "RS 93 · 2nd of 21 in Energy · sector RISING". Sectors are scored on the **median** member rank, not the mean, so one runaway constituent cannot carry a weak sector; groups with fewer than three ranked members get a `null` percentile rather than a bottom rank.

4.9 Market breadth

$$\text{breadth} = \frac{\#\{\,i : P_n^{(i)} > \text{SMA}_{200}^{(i)}\,\}}{\#\{\,i : \text{at least 200 bars}\,\}}, \qquad \text{healthy} \iff \text{breadth} \geq 0.5$$

This is the first market-direction gate the engine has had. Minervini's funnel is market → group → stock → base → pivot; only the last three were ever implemented, which is why the original at-pivot measurement pooled corrections with uptrends.

Breadth rather than "the index above its 200-day" for a concrete reason: `^GSPC` and `^VIX` have **zero rows** in both stores. `MarketRegimeService` fetches them live, which serves a Telegram line but cannot be evaluated point-in-time in a backtest. Breadth is computable for any historical date with no lookahead and no extra feed, which is what makes the gate *testable rather than merely assertable*. It is arguably the better measure anyway: an index can be carried above its average by a handful of mega-caps while most stocks are below theirs, and it is the latter that decides whether a random breakout has support.

The 0.5 threshold is deliberately the natural dividing line rather than a fitted one. With ~11 gated breakout events a year, tuning a threshold against that sample would be pure overfitting.

4.10 Fundamentals overlay

Advisory context, never a gate. A 0–100 blend of valuation, quality and growth from Yahoo's `quoteSummary`. The valuation term is $100 - 2 \cdot \text{forwardPE}$, clamped at 0 — so anything above 50× scores zero on it.

THE GBP TRAP, WHICH WAS LIVE

The LSE quotes in pence and Yahoo labels the currency `GBp`, but does not apply that consistently. `forwardPE` comes back as pence-over-pounds — price in GBp divided by EPS in GBP — inflating the ratio 100×. `AZN.L` reports 1065 for a real 10.7; the median UK forward P/E across the mid-cap pool read **1,190**. Every GBp-quoted name therefore scored zero on valuation.

`normalizeForwardPE` rescales only inflated values and discards anything landing implausibly cheap — a blanket divide would have corrupted `UKW.L` (9.5) and `BBOX.L` (16.8), which arrive correctly in pounds. Currency decides, not the `.L` suffix: `MTLN.L` lists in London and quotes in euros.

A second GBp trap is recorded but has no reader: `marketCap` is in *pounds* while the price is in *pence*. `HSBA.L` reports 262.7e9 against a 1533.2 GBp close — 17.1bn shares, which is right; 262.7e9 pence would be a £2.6bn HSBC.

4.11 Commission

Nordnet charges, *per order*:

$$c(V) = F + \pi V$$

Both terms always apply. On the Mini class this account uses, $\pi = 0.25\%$ and F is 9 SEK on non-Nordic venues, 1 SEK on Nordic ones. Buy and sell are separate orders, so a round trip is two charges. The rate card labels F "Minst", which reads as a floor — that misreading produced a wrong model that stood for five days.

Because F is always present, the round-trip *rate* falls monotonically with size toward, but never reaching, $2\pi = 0.50\%$:

Computed from the compiled `nordnet-fees.ts` at USDSEK 9.4645. Not transcribed.

Position	Per order	Round trip	% of position
$100	$1.201	$2.402	2.402%
$250	$1.576	$3.152	1.261%
$400	$1.951	$3.901	0.975%
$550	$2.326	$4.651	0.846%
$1,000	$3.451	$6.902	0.690%
$2,000	$5.951	$11.902	0.595%
$5,000	$13.451	$26.902	0.538%

The reference point is where the two components cost the same, $F/\pi = 3{,}600$ SEK ≈ **$380**. Below it the bill is mostly the fixed fee and size helps sharply; above it the percentage dominates and size barely moves the rate. `SIGNAL_MIN_POSITION_USD = 400` sits just past that point, where the round trip first drops under 1% of the position.

Each extra leg in a basket is another order and therefore another fixed 9 SEK. That makes the basket-trimming loop in `StrategiesService.sizeBasket` load-bearing at *all* sizes, not only for small slices.

4.12 The pre-buy screen

Advisory only, by design, and attached to every YAHOO BUY that fires. Each line is best-effort and omitted when unavailable, and it is fetched *only* for the handful of symbols that actually fired — never the whole watchlist, since the free tier allows 60 calls a minute.

Field	Computed how
`trend200d`	in-house, from the signal's own snapshot
`sectorTailwind`	in-house — median 3-month return of the peer group, ±3%
`analystTrend`, `epsRevisionTrend`	fetched
`nextEarningsDate`, `daysToEarnings`	fetched
`headlines`	fetched, capped

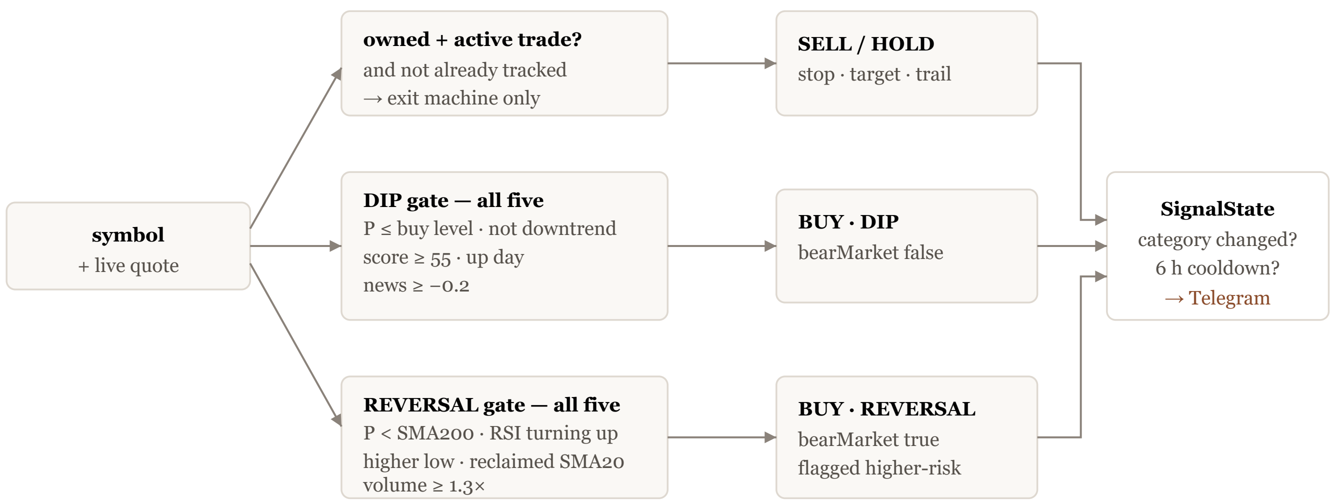
The peer group is the provider's sector where known, falling back to the curated theme category. Sector is the better grouping precisely because it is *coarser*: the curated taxonomy produced peer groups of one or two names, and a median over two names is noise. Real Estate went from 2 names to 29 and Utilities from 6 to 36 when the switch was made.

4.13 News sentiment

A *risk gate and size modifier*, never a standalone directional signal. A DIP BUY requires `newsScore ≥ -0.2`, skipped entirely when no news data is available, and a negative-but-passing score shrinks the suggested size by $\max(0.25,\ 1 + \text{score})$. It cannot create a signal; it can only suppress or shrink one. In the current log `newsScore` is null throughout — the provider is unconfigured, so the gate is inert.

5 · The decision rules

Every symbol is classified on every pass, in its native currency. There are two entry paths, one exit machine, and three screens that run beside them rather than inside them.



Classification. The exit machine and the two entry paths are mutually exclusive per symbol per pass.

5.1 DIP — the primary buy path

All five conditions must hold:

1. $P_{\text{live}} \le$ the adaptive buy level (§4.3) — the dip itself.
2. **Not** in a confirmed downtrend: $\neg(\text{SMA}_{50} < \text{SMA}_{200} \wedge P < \text{SMA}_{200})$. Avoid the falling knife.

3. Composite score ≥ 55. The score now *gates*; it used only to rank.
4. **Up day**: $P_{\text{live}} > P_n$. Reversal confirmation — do not buy while it is still falling.
5. News sentiment ≥ −0.2, skipped when unavailable.

Condition 4 is the cheapest and most underrated of the five. It costs nothing, requires no extra data, and converts "price is low" into "price is low and has stopped going down today".

The emitted signal carries `signalType: 'DIP'`, `bearMarket: false`, and a reason that names which factors aligned. 83 DIP signals are in the log, spanning 2026-06-17 to 2026-08-24.

5.2 REVERSAL — the controlled way to buy weakness

The DIP path is uptrend-only by construction: conditions 2–4 block any name trading below its 200-day, which is exactly the beaten-down set that never produced a buy. Catching those is catching a falling knife *unless the bottom is confirmed*. REVERSAL requires a confirmed turn, not merely a low price, and is always flagged as higher-risk.

#	Confirmation	Test	Why
1	beaten down	$P < \text{SMA}_{200}$	the DIP path would have rejected it
2	RSI turning up, not overbought	$\text{RSI}_n > \text{RSI}_{n-1}$ and $\text{RSI}_n < 70$	the washout happened *during* the decline; we want RSI rising back through mid-range, not a vertical rip that is already extended
3	higher low	$\min(P_{n-4..n}) > \min(P_{n-9..n-5})$	the structural sign that new lows have stopped
4	reclaimed the 20-day	$P \geq \text{SMA}_{20}$	price has retaken the short-term average
5	capitulation volume	$V_n \geq 1.3\,\overline{V}_{20}$	real bottoms turn on volume; a quiet drift up is not confirmation

Condition 5 is skipped when volume data is unavailable, and the reason line says "volume n/a" rather than silently passing. The volume fetch is deliberately lazy: the cheap close-based structure check runs first, and only when it holds does the engine spend a request.

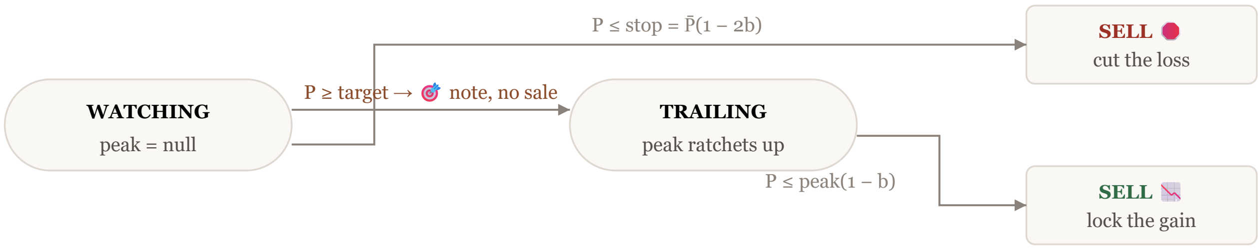
REVERSAL sets `adaptiveLevel` to SMA_{20} rather than to a buy level — the same database column carrying a different quantity depending on path. Worth knowing before reading the log.

Note what REVERSAL does *not* require: a composite score above 55. The two audited REVERSAL signals both scored 35. That is coherent — the score is a dip-attractiveness measure, and a name that has just reclaimed its 20-day on rising RSI is by construction no longer at the lower Bollinger band — but it means the two paths apply materially different standards of evidence, and only one of them has been measured against a base rate.

5.3 The exit machine

Only positions explicitly flagged `isActiveTrade` are managed. **Core holdings are never sold by the engine.** A position with a live tracked trade is also skipped, because that system already owns the exit, anchored to the real fill price; without the guard both would watch the same holding with different levels and each raise its own SELL.

Two phases, persisted in `SignalState.trailingPeak` — `null` means watching, a number means trailing.



Touching the target does not sell. It records the peak and switches to trailing, so winners are not capped.

Every SELL reason reports the **net** gain after the round-trip commission — the number that decides whether the trade was worth doing, not the gross one.

`SIGNAL_EXIT_MODE` selects between this `trailing` machine and `hold-with-stop`, which drops the take-profit entirely in favour of a single wide 22% peak-trailing stop. **The constant is set to `trailing`, and the constant is what runs.** This is worth stating plainly because an earlier revision of the operations guide claimed `hold-with-stop` had been made the default; it had not. Both were measured across 153 names: hold-with-stop improved the average edge from −23.5pp to −20.8pp by not capping winners, and *neither beat buy-and-hold*. The lesson recorded at the time still stands — the exit was never the main culprit.

5.4 LEADER — the VCP breakout alert

Fires at 22:40 on weekdays, after the US close so the day's bar and its breakout volume are final rather than intraday. Two rules govern what actually gets sent, and both exist because of a specific observed failure:

- **Breakouts only.** `AT_PIVOT` is a *state*, not an event — a name can hold its pivot for a fortnight, so alerting on it re-sent the same two dozen names nightly. A breakout is dated and self-limiting: roughly four a day across 833 names.
- **A five-day per-symbol cooldown.** The breakout bar keeps satisfying the test for several sessions afterwards, until its volume surge rolls out of the 50-day average. Without the cooldown one event alerts every evening for a week.

No RS floor is applied on top, because every candidate reaching this point has passed all eight Trend Template criteria and criterion 8 *is* RS ≥ 70. At-pivot and forming names remain visible in `GET /signals/leaders`, the watchlist columns and the Trend tab — they simply do not page you.

The alert also writes a `SignalLog` BUY row with `signalType: LEADER` and the screen stop, so the Simulation tab grows a Leader curve. It is **forward-only** — nothing is backfilled, because a breakout that was never alerted was never a signal anyone could have acted on, and inventing its history would make the curve a backtest wearing a live-results label.

5.5 TT8 entrants — the daily quality alert

Fires weekday mornings at 07:00 on names that *crossed into* 8/8 on the settled US close, gated at RS ≥ 90, with a 10-day per-symbol cooldown. The gate is a measurement, not a preference:

New 8/8 entrants per day over 94 trading days across 752 names.

Population	mean	median	p90	max
all 8/8	13.4	7	22	142
RS ≥ 90 only	3.9	2	6	58

An ungated alert averages 13 names and spikes to 142 when the cross-section re-ranks — unreadable within a week. Gated it is a median of 2 and silent on 28 of 94 days. The 10-day cooldown stops a name hovering on a criterion boundary re-alerting every other day.

Crucially, TT8 rows are written under `category: 'WATCH'`, which the simulation never reads — it selects `category IN ('BUY','SELL')`. Auto-logging them as trades would be roughly a thousand lots a year: an equal-weight index of the screen rather than a strategy, swamping every other curve while representing nothing anyone traded. The TT8 curve appears only once a real purchase is recorded.

The engine held 47 TT8 rows at the time of audit, the most recent five written on 2026-08-27 at 07:00 CET. Every one carries the reason `Entered Trend Template 8/8 at RS nn` and nothing else — no score, no stop, no target. That is correct for a watch item.

5.6 Shortlist and 5.7 the EV ranking

The **shortlist digest** is the same 8/8 ∩ RS ≥ 90 population sent as a standing list on the 1st and 15th, capped at `SIGNAL_SHORTLIST_MAX = 12` with the overflow summarised rather than silently truncated. It answers "what is currently strong", where the entrant alert answers "what changed".

The **EV ranking** in `StrategiesService` is what turns candidates into an allocation. Eligibility is score ≥ 45, not in a downtrend unless a confirmed reversal, not recently exited (14 days), and annualised volatility ≤ 90%; survivors are ordered by expected value, sized by `sizeBasket`, and trimmed until the fee ratio clears `SIGNAL_STRATEGY_MAX_FEE_RATIO = 0.1`. Redundant picks — correlation above 0.4 — are penalised by half.

Two properties of `sizeBasket` are worth recording. It reconstructs each leg's principal as `shares × price`, which is exact by construction. It previously used `cost - fee`, and once the commission became fractional that landed a fraction of a cent low (120.95 − 0.95092 = 119.99908), so the next `floor()` silently dropped a whole share and the greedy remainder-minimisation pass stopped running *at all*. A latent bug that only surfaced when an unrelated model changed.

6 · Delivery — how a signal reaches the phone

6.1 The path

Cron → Bull queue → processor → `SignalsService` → classification → `SignalState` diff → cooldown gate → format → Telegram. Telegram is the only push channel; the PWA reads `GET /api/v1/signals` and pulls.

6.2 De-duplication

Telegram fires **only when a symbol's category changes** against the stored `SignalState.lastSignal`, and only if `SIGNAL_NOTIFICATION_COOLDOWN` (6 hours) has elapsed. Running the job twice with no change sends nothing. This is why the 30-minute cadence does not translate into 48 messages a day.

The screens carry their own, longer cooldowns keyed in the same table: `LEADER:<dataSource>:<symbol>` at 5 days, TT8 at 10.

6.3 Message anatomy

A DIP BUY, as constructed by `formatMessage` — this is the exact shape of the DVA message reconstructed in §7.1:

```
📊 *Trading signals* (2026-08-24)

🟢 *BUY* DaVita Inc. (DVA)
Down 27.7% off its 30-day high (buy zone <= 176.39 USD); score 75/100;
up 0.2% today; fundamentals 84/100.
Price: 174.22 USD
~33% chance of hitting target in 252 trading days
Suggested: 207.5
```

Line	Field	Note
header	emoji + label + name + symbol	REVERSAL gets `⚠️ REVERSAL BUY (bear-market)`; an ETF dip gets its own label because its buy zone is calibrated differently
reason	the aligned factors	drop off high, buy zone, score, up-day %, then news and fundamentals if present
price	`livePrice` + currency	native currency, correctly labelled
probability	`hitTargetProbability`	rounded to a whole percent, over **252** trading days — a different horizon from the 42 that sized the levels
size	`suggestedAmount`	**no currency label** — Finding 3

The leader alert uses a denser three-line form and is sent as Telegram **HTML** rather than legacy Markdown. That is not cosmetic: Markdown mis-parses `_` inside a URL, and TradingView writes Nordic share classes with one (`ASSA-B.ST` → `OMXSTO-ASSA_B`), so inline links would have broken on precisely the Nordic names. `sendMessage` takes an optional parse mode defaulting to Markdown, leaving every other caller unaffected.

```
🚀 UNP 303.97 · 8/8 · RS 79
pivot 298.15 +2.0% · vol 1.45× · dry 0.73× · stop 281.17
Yahoo · TradingView · Perplexity
```

The full scorecard — all eight criteria with the number each was decided on, both volume ratios, the contraction sequence, the base length — lives in the ticker dialog's Trend tab and in `GET /signals/leaders`. Repeating it in the message buried the two facts that decide whether to open the chart at all.

6.4 What is persisted

Every delivered signal writes a `SignalLog` row: the indicator snapshot, the levels, the probability, the EV, the reason string, and a JSON `metrics` column carrying the asset sub-class and the pre-buy screen exactly as it read at fire time. That column is what makes this audit possible without a schema migration.

Two fields deserve a note. `conviction` is deliberately left unwritten — the metric drove no decision anywhere and its `50 + EV × 1000` rendering was unbounded and misleading; the column survives so historical rows stay readable. And `stopLoss`/`takeProfit` on a BUY row are computed as $P_{\text{live}}(1 \mp \cdot)$ **without** the fee adjustment the exit machine applies — see §10.4.

The signal corpus at the time of audit.

Type	Category	n	First	Last
DIP	BUY	83	2026-06-17	2026-08-24
REVERSAL	BUY	64	2026-06-17	2026-08-25
TT8	WATCH	47	2026-08-24	2026-08-27
LEADER	BUY	4	2026-08-24	2026-08-24
—	REINVEST	1	2026-06-22	2026-06-22
—	HOLD	1	2026-08-21	2026-08-21

Two things stand out.

First, **there is not a single SELL row in the corpus** — and this is not because the exit machine is idle. Six of the eight configured symbols are flagged `isActiveTrade`, and five `SignalState` rows carry a non-null `trailingPeak`, meaning five positions have touched their take-profit and switched into the trailing phase. None has since fallen a full band below its running peak. That is entirely consistent with a rising market over the logged window, but it has a consequence worth stating plainly: **the SELL branch of the exit machine has never executed in production.** It is covered by unit tests and by the backtest harness; it has never actually sold anything. The stop-loss branch is likewise untested live.

Second, the LEADER path produced four signals on one day and none since — the expected shape given a five-day per-symbol cooldown and roughly eleven gated events a year.

7 · Worked examples — real signals, recomputed

Each example takes a signal the engine actually sent, reconstructs the inputs it saw, re-runs its own compiled services, and diffs every stored field. Where the reproduction fails, the failure is the finding.

7.1 DVA — a DIP signal, reconciled exactly

DaVita Inc., 2026-08-24 at 14:00:44 UTC. Stored as `SignalLog` id `292ec958…`, category BUY, `signalType: DIP`.

Step 0 — the series

The engine fetched 520 calendar days of `MarketData` and weekday-filtered it. Reconstructing that window with a `createdAt <= 2026-08-24T14:00:44Z` cut-off yields **356 closes** ending at 173.82 (Friday 21 August).

That series reproduces the stored RSI to seven significant figures — and gets Bollinger %B wrong by 7.8% and the MACD histogram by 34.7%. Since all three come from the same array in the same function call, the array must be wrong in a way RSI cannot see. §4.1 gives the mechanism: **Wilder's RSI is exactly invariant to an appended zero-change bar, and Bollinger and MACD are not.**

Appending one duplicate of the final close — 357 bars ending 173.82, 173.82 — reproduces everything:

Reconciliation. "Computed" is produced by loading the engine's own compiled `IndicatorsService` *and* `ForecastService`.

Field	Stored	Computed	Δ	
score	75	75	0.00000%	✓
rsi	24.228792149	24.228795889	0.00002%	✓
bollingerPctB	0.2867298109	0.2867298736	0.00002%	✓
macdHistogram	−1.5954394650	−1.5954389976	0.00003%	✓
adaptiveLevel	176.39339331	176.39339332	0.00000%	✓
stopLoss	111.97567135	111.97567136	0.00000%	✓
takeProfit	220.90324649	220.90324648	0.00000%	✓
reachProbability	0.3343661971	0.3343661971	0.00000%	✓
expectedValue	−0.1482185145	−0.1482185144	0.00000%	✓
forecastLower	100.03173819	100.03173819	0.00000%	✓
forecastUpper	303.42978088	303.42978088	0.00000%	✓

Eleven of eleven. The residual 2×10^{-5}% on the three snapshot fields is `float8` round-tripping through Postgres, not a difference in the mathematics.

WHAT THE RECONCILIATION PROVES — FINDING 2

The duplicated tail bar is not an artefact of the reconstruction; it is *required* to reproduce what the engine wrote. The hourly gatherer had inserted a row dated 2026-08-24 at 13:30:56 UTC — thirty minutes before the signal fired — initially carrying the previous close of 173.82. That row has since been revised to the settled close of 176.50, which is why neither a naïve `<` nor a naïve `<=` cut-off reconstructs it.

So **every intra-session evaluation runs on a series whose last two entries are the same number.** Bollinger %B sees 19 real sessions instead of 20; MACD's short EMA absorbs a flat day that did not happen. Here that moved %B from 0.2645 to 0.2867 (+8.4%), the MACD histogram from −2.1485 to −1.5954 (25.7% closer to zero), and the composite score from 76 to 75.

Step 1 — the indicators

From those 357 closes: RSI $= 24.23$ (deeply oversold), $\%B = 0.287$ (in the lower third of the band), MACD histogram $= -1.595$ (negative), $\text{SMA}_{200} = 161.47$ against a live price of 174.22 (above), 12-month momentum $= +26.3\%$ (positive), and $\sigma = 2.7559\%$ per day.

Step 2 — the composite score

$$S = \frac{0.25(100 - 24.2288) + 0.30(1 - 0.2867)100 + 0.15(30) + 0.15(100) + 0.15(100)}{1.00}$$

Term	Input	Contribution
RSI, 0.25	$100 - 24.2288 = 75.7712$	18.943
Bollinger, 0.30	$(1 - 0.28673) \times 100 = 71.327$	21.398
MACD, 0.15	hist < 0 → 30	4.500
Trend, 0.15	$174.22 \geq 161.47 \to 100$	15.000
Momentum, 0.15	$+26.3\% \geq 0 \to 100$	15.000
total, rounded		**74.841 → 75**

Note the shape of this: **40.3 of the 74.8 points come from the two weakness terms.** DVA scores highly because it has fallen hard, and passes the trend and momentum gates because it had risen for a year before that. That is precisely the profile the DIP path is built to find — and precisely why the score cannot also serve the Trend Template.

Step 3 — the horizon band and the four levels

$$b = \sigma\sqrt{H} = 0.027559 \times \sqrt{42} = 0.027559 \times 6.48074 = 0.178623$$

The 30-day high is 240.96, giving a drop of $1 - 174.22/240.96 = 27.7\%$ — the figure in the message. The buy level takes the more conservative reference:

$$\text{fixed} = 240.96(1 - 0.10) = 216.86, \qquad \text{sigma} = 240.96\,(1 - 1.5 \times 0.178623) = 176.39$$

The sigma leg binds — DVA is volatile enough that a 10% dip would be noise — so the buy zone is **176.39**. Live price 174.22 is below it. Condition 1 holds.

$$g = \max(0.134,\ 1.5 \times 0.178623) = \max(0.134,\ 0.26793) = 0.26793$$

$$s = 2 \times 0.178623 = 0.357246$$

$$\text{takeProfit} = 174.22 \times 1.26793 = 220.903, \qquad \text{stopLoss} = 174.22 \times 0.642754 = 111.976$$

Both match the stored values exactly. Note that these logged levels carry **no fee adjustment** — the $+f$ term of §4.3 is zero here because `toSignalLogInput` derives them from `livePrice` and the raw percentages, not from the exit machine. See §10.4.

Step 4 — the probability

EWMA volatility over the same returns is $\sigma_{\text{ewma}} = 3.4956\%$ per day — **1.27× the sample** σ, because RiskMetrics' 16-day memory is still full of the late-July collapse from 240 to 180. Over the 252-day forecast horizon with drift set to zero:

$$z = \frac{\ln(1+g)}{\sigma_{\text{ewma}}\sqrt{252}} = \frac{\ln 1.26793}{0.034956 \times 15.8745} = \frac{0.237424}{0.554889} = 0.42788$$

$$\Pr[\,P_{252} \geq 220.90\,] = 1 - \Phi(0.42788) = 0.33437$$

Stored: 0.334366197. The forecast band is the same quantities at $k = 1$:

$$\left[\,174.22\,e^{-0.554889},\ 174.22\,e^{+0.554889}\,\right] = [\,100.03,\ 303.43\,]$$

Stored: 100.032, 303.430. Both exact.

Step 5 — expected value, and how to read it

$$\text{EV} = 0.334366 \times 0.267935 - (1 - 0.334366) \times 0.357246 = 0.089585 - 0.237804 = -0.148219$$

Stored: −0.148219. The break-even probability for this payoff geometry is $s/(g+s) = 0.357246/0.625181 = 57.1\%$, against an actual 33.4%. **This is a negative-EV trade by the engine's own arithmetic, and the engine sent it anyway.**

That is not a contradiction — it is the design. EV is a *ranking* quantity used by the strategy builder to order candidates when cash arrives; the DIP path's own gate is the five conditions of §5.1, which say nothing about EV. Under zero drift a $+1.5\sigma$ terminal move is structurally improbable, so essentially every candidate is negative-EV and the ordering is what carries the information (§4.5). But it does mean a reader who takes "EV = −0.148" at face value would conclude the engine recommends trades it expects to lose 14.8% on, and the message does not carry the EV at all — which, given the above, is probably the right call.

Step 6 — size and cost

$$\sigma_{\text{ann}} = 0.027559 \times \sqrt{252} = 43.76\%, \qquad \min\left(0.25, \frac{0.20}{0.4376}\right) = \min(0.25,\ 0.4571) = 0.25$$

The cap binds. With a cash balance of $830.00, the suggestion is **$207.50** — matching the stored value exactly, and confirming the balance. On a $207.50 position the Nordnet Mini round trip is $1.470 + $1.470 = **$2.94, or 1.42% of the position**. Against a 26.79% target that is 5.3% of the intended gain; against the 35.72% stop it is 4.0% of the risk. The position sits below `SIGNAL_MIN_POSITION_USD` = 400, where the round-trip rate is still climbing steeply.

Step 7 — the five gates

#	Condition	Value	
1	$P \le$ buy level	174.22 ≤ 176.39	**pass**
2	not a downtrend	price 174.22 > SMA200 161.47	**pass**
3	score ≥ 55	75	**pass**
4	up day	174.22 > 173.82, +0.230%	**pass**
5	news ≥ −0.2	no data → skipped	**inert**

Gate 4 passes by 0.23% — 40 cents. Had DVA opened four cents lower, no signal. That is the intended behaviour of a reversal-confirmation gate, but it is worth seeing how thin the margin can be, and worth noting that the duplicated tail bar (Finding 2) is what fixed `previousClose` at 173.82 in the first place.

The pre-buy screen, stored verbatim in `metrics`: above the 200-day, sector RISING, analyst trend IMPROVING, EPS revisions FLAT, next earnings in 64 days, one headline. Fundamentals scored 84/100. Nothing there argued against the entry.

7.2 PGHN.SW — a REVERSAL, and an instructive near-miss

Partners Group Holding AG, 2026-08-25 at 22:00:31 UTC, quoted in CHF. Score 35 — a number that would fail the DIP gate outright.

The five REVERSAL confirmations, as the message reported them.

#	Confirmation	Value
1	beaten down	726.00 vs SMA200 855.18 — 15.1% below
2	RSI turning up, < 70	55.5, rising
3	higher low	confirmed over the 5/5 window
4	reclaimed SMA20	726.00 ≥ 714.51
5	capitulation volume	1.8× the 20-day average

The bands reconcile to four decimal places — $b = 0.13671$, stop 527.53, target 874.85, both within 0.006% of stored. The reason line's own numbers are internally consistent: down 1.8% off the 30-day high of 739.40, RSI 56, volume 1.8×.

But four fields do **not** reconcile, and the reason is worth recording as a limit on this method:

Field	Stored	Computed	Δ
bollingerPctB	0.600027	0.613591	2.26%
macdHistogram	−0.395809	−0.361860	8.58%
reachProbability	0.229136	0.236004	3.00%
forecastUpper	933.565	940.968	0.79%

Two causes, and neither is an engine defect. First, this signal fired at 22:00 CET, after the Swiss close, so its same-day row already held a settled price that has since been revised — the reconstruction cannot recover the intermediate value the way it can for a mid-session US signal. Second, both %B and the EWMA-driven fields are sensitive to the exact first bar of the 520-day window, and `subDays(new Date(), 520)` anchors on the run *instant*, which is not recoverable from a row timestamp alone.

The contrast with §7.1 is the useful part. **Quantities driven by the pinned 252-day σ reproduce to four decimals; quantities driven by seeded recursions — EWMA, the EMAs, and any 20-bar window — reproduce only to within a few percent.** That is a property of the estimators, not of the audit, and it is worth knowing before anyone tries to reconcile a historical signal by hand.

Finally, this signal is Finding 3 in the wild. Its message read `Price: 726 CHF` and, on the next line, `Suggested: 66.85`. The intended instruction was 66.85 *dollars* — about 0.08 shares. Read as CHF it is 0.09 shares; the error is small here because CHF and USD are near parity. The same 66.85 went out for `SAGA-B.ST` at 169.50 SEK, where the gap is 9.5×, and one REVERSAL signal in the log is on a `GBp`-quoted name, where reading the number as pence would be off by more than a hundredfold.

7.3 MRK — the only LEADER breakouts the engine has ever sent

On 2026-08-24 at 13:45 UTC, four VCP breakouts fired — the entire LEADER corpus.

Symbol	Price	Pivot	Above	Vol	Dry-up	RS	Stop
MRK	151.19	131.09	+15.3%	1.53×	0.78	93	139.85
CF	130.11	120.55	+7.9%	1.51×	0.69	88	120.35
RVTY	121.97	—	—	1.53×	0.62	88	—
UNP	—	298.15	+2.0%	1.45×	0.73	79	—

The stop reconciles trivially: $151.19 \times (1 - 0.075) = 139.851$, exactly `SIGNAL_LEADER_STOP_PCT` below the entry. The suggested size is a flat \$550 — `SIGNAL_BACKTEST_POSITION_SIZE`, not the volatility-targeted figure the DIP path uses. Two entry paths, two unrelated sizing rules, and only one of them responds to the name's volatility.

> **THESE FOUR ALERTS WERE FIRED BY A DETECTOR THAT WAS REPLACED THE SAME DAY**
>
> MRK at +15.3% above its pivot is not a breakout anyone should buy — Minervini's own rule is that a name more than a few percent past the pivot is extended. Under the corrected detector shipped on 2026-08-24, **all four of these are rejected**: MRK for "only 2 consecutive tightening contractions out of 9 pullbacks", UNP for "only 1 out of 6", CF for "base is only 14 days".
>
> These rows remain in the log and remain on the Simulation tab's Leader curve. They are a record of what the engine sent, which is what a log is for — but the curve they feed is built on structure that was not there.

7.4 UNP versus JNJ — why volume is two tests

The clearest illustration in the corpus, both names on 2026-08-20, both 8/8 on the Trend Template, both with three clean contractions, both with price through the pivot.

	UNP	JNJ
contraction sequence	7.5% → 3.5% → 2.6%	5.2% → 4.0% → 3.4%
price vs pivot	+1.95% (pivot 298.15)	+1.22% (pivot 264.15)
dry-up ratio	0.73×	**0.65×**
breakout volume	**1.45×**	0.80×
status	BREAKOUT	AT_PIVOT

JNJ's supply dried up *more* thoroughly than UNP's — 0.65 against 0.73 — and it still did not confirm. It crossed its pivot on **below-average** volume: fewer shares changed hands on its breakout day than on an ordinary day, so nothing had to be bought to lift it through the level, and no accumulated position is defending it.

Collapse the two ratios into one number and this distinction disappears. That is why they are reported as a pair. Note also that under the corrected status logic JNJ, being *above* its pivot without volume, is a `FAILED_BREAKOUT` rather than `AT_PIVOT` — the one-sided fix of §4.7 changes exactly this case.

7.5 Fees, worked

Two shares of NVIDIA at about \$212.50, a \$425 order, non-Nordic, Mini class, USDSEK 9.4645:

$$c = \frac{9}{9.4645} + 0.0025 \times 425 = 0.9509 + 1.0625 = \$2.013$$

and the user's own reference case, a \$100 order:

$$c = 0.9509 + 0.25 = \$1.2009$$

Both computed by calling the compiled `nordnetCommissionUsd`, not by transcribing a table. The additive structure is the whole point: at \$100 the fixed component is 79% of the bill, at \$5,000 it is 7%.

A worked comparison that follows directly, using the audited signal's own sizing. A four-leg basket of \$207.50 positions pays $4 \times 2 \times \$1.470 = \$11.76$ in commission on \$830 deployed — **1.42%**. The same \$830 in a single position pays 2 × \$3.026 = \$6.05 — **0.73%**. Diversifying across four names costs 69 basis points of round-trip friction, every time. Whether that buys enough diversification is a portfolio question the engine does not answer; the `sizeBasket` trimming loop (§5.7) is what stops it from being answered badly.

8 · Decision history

Seven commits span 2026-07-10 to 2026-08-05. Everything after that date is uncommitted working-tree work, documented in the operations guide's dated sections. Both sources are used below; no date here is asserted from memory.

The timeline. "Effect" is the measured or structural consequence for what the engine sends.

Date	Decision	Effect on signals
2026-07-02	`adaptiveBuyLevel` takes `horizonDays` explicitly instead of hard-coding a 20-day window	The buy trigger became consistent with the stop/target/trail, which already used H. Before this the entry band and the exit bands were sized on different horizons.
2026-07-08	ETF buy calibration: $d = 5\%$, $k = 1.0$	Unblocked an entire asset class. **No ETF signal had ever fired** — a diversified ETF does not print a 10% drop off a 30-day high outside a crash, and the 1.5σ leg demanded 20–40% of thematic ETFs.
2026-07-09	Allocation moves 67/33 → **60/40**; the 750 USD monthly playbook is adopted; a plan cron is added for the 25th	Not a signal change — a *capital* change. The risk sleeve grows by a fifth, so every signal now sizes against a larger base.
2026-07-10	**The engine ships.** Indicators, forecast, backtest, the DIP and REVERSAL paths, the exit machine, Telegram, Academy, Simulation	Everything in §4 and §5 except the Minervini layer dates from here.
2026-07-12	Fund/ETF detail views, correlation matrix, overlap-aware fund ranking	Introduced the 0.4 redundancy threshold that still penalises correlated picks in the EV ranking.
2026-07-14	Pre-buy screen attached to fired BUYs — advisory only	Context without a gate. Deliberately fetched for the handful of symbols that fired, never the watchlist.
2026-08-03	Frozen signal-target tracking for real trades	A real fill now owns its own exit, anchored to the actual price. This is why `evaluateSymbol` skips a position with a live tracked trade — otherwise two systems would watch one holding with different levels.
2026-08-04	Simulation performance chart; per-signal-type FIFO buckets	Made it possible to see whether a path works. Non-DIP types match in isolated buckets so one strategy's exit can never close another's entry.
2026-08-05	`toTradingDayCloses` — the calendar-day/trading-day fix	The single most consequential correction in the history. Before it every window spanned ~5/7 of the sessions its name claimed and σ was deflated by ~15.5%, propagating into every stop, target and probability. Finding 1 is this same failure returning through a different door.
2026-08-21	The Minervini layer lands: `OhlcBar`, Trend Template, VCP, cross-sectional RS. Commission model corrected from a flat "$5/side" to `max(pct·V, min)`	The engine gains a second, opposite-signed way of looking at stocks. The fee model becomes wrong in a new way (§8.1).
2026-08-22	Both volume ratios surfaced, not one	The half of the VCP test that *gates the whole pattern* had been invisible in the UI, the API and Telegram.
2026-08-23	Universe widened to full S&P 500 + EURO STOXX 50 membership; 424 profiles created; sectors backfilled	RS percentiles become meaningful over ~750 names rather than 334. Peer groups go from as small as one name to 29–123. **And the importer plants Finding 1.**
2026-08-24	VCP detector rewritten; `AT_PIVOT` made one-sided; market breadth added; TT8 entrant alert and shortlist digest added; the leader alert's three faults fixed	Valid-VCP share falls from 38.4% to 2.5%. The alert fires for the first time in its life. The Minervini funnel gains its missing first step.
2026-08-25	Nightly OHLC gather; boot catch-up sequenced; 100 curated European mid-caps	The bars stop being frozen. Every high/low/volume metric becomes current for the first time since the backfill scripts were last run by hand.

Date	Decision	Effect on signals
2026-08-26	Commission corrected to **additive**; `feeToRiskRatio` and `SIGNAL_MAX_FEE_TO_RISK_RATIO` removed	The fee rate stops being flat above $380 and starts falling with size. Every doc and lesson teaching "trading larger buys nothing" is inverted.

8.1 The commission model was wrong twice, in opposite directions

Worth isolating, because it is the clearest case in the history of a wrong model producing confidently wrong *advice* rather than merely wrong numbers.

Period	Model	What it implied
until 2026-08-21	flat $5 per side	fee independent of size; a $300 order costs 3.3% round trip
21 → 26 Aug	$\max(\pi V,\, F)$	rate **constant** above ~$380 — so "trade bigger to amortise the fee" is *false*
from 2026-08-26	$F + \pi V$	rate falls monotonically toward 0.50% — so it is *true*, with diminishing returns

The middle model was not a typo. It was a defensible reading of a rate card that labels the fixed component "Minst" — Swedish for "at least", which reads as a floor. It was also reinforced by a second error: the seven fee-bearing orders in the database all cost 48–49 SEK flat with no percentage on top, and inferring the commission class from them gave the wrong answer twice. Those fills predate a class change; they are the old *Liten* tariff. The operations guide now carries an explicit instruction not to infer the class from them again.

The correction had a latent consequence nobody predicted: it broke `sizeBasket`. See §5.7 — with whole-dollar fees the principal reconstruction was exact; with a fractional commission it landed a fraction of a cent low and silently dropped a share.

9 · Changes of direction, and the philosophy that survived them

9.1 The founding position (2026-07-10)

Buy confirmed dips in names that are otherwise healthy; size them by volatility; define the downside with a stop; let winners run with a trail; rank by expected value with **zero drift**; keep the trading sleeve small because the fund core does the real compounding. Deliberately pessimistic: the system should say "nothing compelling" far more often than "buy".

9.2 The first reversal: buying strength (2026-08-21), and its rejection

The backtest evidence was stark. Across 153 watchlist names, only 29% beat buy-and-hold, mean edge −23.5pp. On the big winners the active rule *missed the run entirely* — ASML, Caterpillar, Dell and Revolution Medicines all returned roughly 0% traded while holding returned +127% / +150% / +194% / +347%. The rule only "won" on names that fell, by cutting the loss.

The intuitive diagnosis — *the entry picks bad tickers because the score rewards weakness* — became the working hypothesis, and the entire Minervini layer was built to test it. **The hypothesis was rejected by its own test.**

Event study against the base rate of every symbol-day. This is the measurement that decided the engine's direction.

Signal	21d	63d	126d
DIP (the existing engine)	**+1.22pp (t=3.07)**	**+3.00pp (t=4.66)**	**+4.35pp (t=4.24)**
Trend Template 8/8	−0.18pp	−0.26pp	**+1.45pp (t=4.11)**
LEADER at pivot	**−0.51pp**	**−1.57pp (t=−3.25)**	−1.20pp
LEADER breakout	−0.86pp	−0.26pp	+1.90pp (t=0.90)

The dip entry beats the base rate at *every* horizon, with win rates of 61/78/79% against a base of 56/61/66%. The Minervini breakout beats it at none.

And the plan to demote DIP to "dips inside confirmed leaders" turned out to be arithmetically impossible: criterion 5 passes on 0.3% of dip days, and the combination produces zero signals (§4.6).

What changed, therefore, was not the buy signal but the status of the new work. DIP stayed primary and unmodified. The leader screen shipped as a research surface — an endpoint, two watchlist columns, a Telegram report — labelled unproven everywhere it appears. §0.3's diagnosis was refined rather than overturned: the shortfall against buy-and-hold is about *exposure and capped winners*, not ticker selection. The strategy is in the market 4–8% of the time and the benchmark is in it 100%; that gap alone produces most of the headline deficit.

> **THIS IS THE MOST CREDITABLE EPISODE IN THE ENGINE'S HISTORY**
>
> A large, expensive, intellectually attractive piece of work was built specifically to test a hypothesis, the test failed, and the result was honoured rather than argued around. The engine's direction did not change; the *confidence* attached to each part of it did.

9.3 The second reversal: auditing the auditor (2026-08-24)

Three days later the VCP detector itself was audited, and it had been fitting noise. `selectContractionChain` ran a longest-decreasing-subsequence search over every ZigZag swing, keeping any subset that shrank and made higher lows. Across all 757 symbols with bar history:

	before	after
symbols showing a "valid VCP"	291 (38.4%)	2.5%
contraction joins that skipped bars	770 of 770	—
chains where skipped price broke *below* the prior trough	224 of 291 (77%)	0 of 44
chains discarding a pullback deeper than the one kept	281 of 291 (96.6%)	—

The third row is the contradiction: the search enforced higher lows on the swings it *kept* while stepping over stretches that made lower lows. With 10–16 raw swings there are thousands of candidate subsequences, and one nearly always looks like a tightening VCP.

Then the harder finding. A controlled A/B — old chain versus new, identical 752 symbols and 259 dates — showed the fix **did not rescue the signal**: the ungated pivot entry underperformed under both detectors, and slightly worse under the corrected one. But that measurement was answering the wrong question, for four separate reasons: it measured an entry Minervini does not take (at-pivot, not the breakout *through* it); it sampled every fifth trading day, missing ~80% of what is a one-day event; it had no exit, so it measured a strategy nobody runs; and it ignored market direction, his first rule.

The corrected audit — every trading day, five years, trend-gated, base rate recomputed under each exit model — produced this:

Signal, 126-day buy-and-hold	n	mean	win	vs base	t
base rate	651,387	9.60%	63.5%	—	—
Trend Template 8/8	116,555	12.09%	66.4%	**+2.49pp**	**24.14**
at pivot (not his entry)	656	8.83%	63.0%	−0.77pp	−0.93
BREAKOUT (his entry)	109	10.55%	59.6%	+0.96pp	0.36
BREAKOUT + healthy + RS ≥ 90	36	22.88%	77.8%	**+13.28pp**	**2.35**

The honest headline: his actual entry has never been adequately measured here. Five years across 752 names produces 109–124 breakout events — far below the pre-registered n ≥ 200 — so nothing in the bottom rows graduates, including the +13.28pp cell, which is one of 63 cells tested and therefore roughly what one expects to clear $t = 2$ by chance.

The direction is still worth recording: the top-RS breakouts in a healthy market are the best cell in the table at a 77.8% win rate, and that is precisely what the method claims. It is a hypothesis the data cannot yet confirm *or* reject — not a refutation.

What was adopted: the DIP path remains the only measured buy signal. Trend Template 8/8 is promoted from curiosity to the basis of a quality shortlist — at +2.49pp over 126 days on n = 116,555 it is the strongest non-DIP result the engine has. The breakout alert keeps firing on a corrected and much rarer detector, still labelled unproven. Nothing is ranked by pivot proximity.

One further finding from that audit deserves its own line, because it contradicts the method's own doctrine in this universe: **the 7.5% flat stop is actively harmful here.** Applied to the base rate itself at 126 days it takes the mean from 9.60% to 4.73% and the median from +6.04% to **−7.62%**; adding a 15% trail makes it worse again. The stop sits inside these names' normal noise, converting a majority of positions into small losses. That is a universe mismatch, not a flaw in his rule — he buys high-ADR growth names at an exact intraday pivot with the stop just under it, which is a different trade. But `SIGNAL_LEADER_STOP_PCT = 0.075` is still what the LEADER path writes (§7.3).

9.4 The third shift: from hand-run to self-maintaining (2026-08-25)

Less glamorous and arguably more important. Until this point the fork's own data store advanced only when someone remembered to run a script, and nothing in the application said so. Three habits were established: **every store gets a writer inside the application**; **every scheduled job gets a boot catch-up**, because a missed gather is a permanent hole where a missed alert is only a missed notification; and **every alert reports how many items it sent**, because before that "nothing qualified today" and "the alert is broken" were indistinguishable from the outside — which is exactly how a permanently-suppressed alert went unnoticed for weeks.

9.5 Where the philosophy stands

Stated plainly, as the code now implements it:

1. **The fund core compounds; the stock sleeve takes risk.** 60/40, and the engine only operates on the 40.
2. **Holding beats trading for names you believe in.** The signals are entry-timing and risk hints for a buy-and-hold sleeve, not a churn engine. This is the backtest's conclusion, and it is why the engine is reluctant by

construction.

3. **No built-in optimism.** Zero drift, terminal rather than touch probabilities, fees modelled on both legs. Negative EV is the normal state.
4. **Buy weakness inside strength (DIP), or confirmed turns (REVERSAL) — and nothing else fires.** Everything Minervini is a shortlist for human judgement.
5. **Volatility-scale everything.** A flat percentage target is a multi-sigma event for a calm stock and noise for a wild one.
6. **Measure before believing, and record the measurement even when it kills the idea.**

Two places where the code still disagrees with that statement are recorded in §10.5 and §10.6.

10 · Audit findings

10.1 Duplicate price rows corrupt every windowed indicator — live

Mechanism. `run-import-index-universe.cjs:149` writes `date: new Date(timestamps[i] * 1000)` — Yahoo's chart timestamp, which is the *session open*: 13:30 UTC for New York, 07:00 or 08:00 for European venues. The hourly gatherer writes UTC midnight. `@@unique([dataSource, date, symbol])` is declared on a `DateTime` column, so the two values never collide and `skipDuplicates: true` has nothing to skip. The same code path writes both stores, so both are affected.

Extent, measured 2026-08-27.

Store	Session-stamped rows	Symbols	Duplicated (symbol, day) pairs	Range
`MarketData`	651,284	527	12,909	2026-07-13 → 08-25
`OhlcBar`	651,284	527	10,639	2026-07-27 → 08-25

426 symbols carry exactly 30 duplicated days each — the S&P/STOXX import cohort — and 100 carry one, the mid-cap cohort imported on 25 August. All 651,284 session-stamped rows were created inside a 60-hour window, 2026-08-23 00:16 to 2026-08-25 09:42.

Impact. Because every indicator window counts array positions rather than dates, over the affected range each one spans about half the sessions its name claims: a 20-period Bollinger reaches back 10 sessions, a 30-day high reaches back 15. The duplicated closes also inject zero returns, which deflates σ.

Measured over 397 affected symbols by computing each snapshot twice — as the engine reads the series, and with the series collapsed to one row per calendar day.

Quantity	Effect of removing the duplicates
composite score	**changes for 364 of 397 symbols**; mean absolute change **5.6 points**, maximum **24**
MACD histogram	**changes sign for 98 of 397** — each worth 10.5 points of composite score
30-day high	**understated for 137 of 397**, median change 0%, worst case CHRW at **+35.3%** (154.75 → 209.42)
daily σ	understated by a median of 3.3%, so every stop, target and trailing band is ~3% too tight
RSI	median change 1.3% — least affected, for the invariance reason in §4.1

The 30-day high row is the one with direct trading consequences. It sets `adaptiveBuyLevel`, so an understated high sets the buy zone too low and **suppresses DIP signals that should have fired**. CHRW is the sharpest illustration and is not hypothetical: it is one of the two DIP signals that fired on 2026-08-24, on a 30-day high of 209.42; the series as it stands today reports 154.75 for the same name.

The composite-score row bites in both directions, because `SIGNAL_BUY_SCORE_MIN = 55` is a hard gate and the mean absolute error is 5.6 points.

Worth noting: 94 of the 12,909 duplicate pairs hold *different* prices for the same day, with a maximum gap of 80 currency units. Those are genuine conflicts, not just repeats.

Direction of a fix — the choice is a real one and belongs to whoever owns the data model, not to this audit. Normalising every ingest path to UTC midnight and de-duplicating the existing rows restores the invariant the unique constraint was meant to express. Keeping session timestamps and de-duplicating in `toTradingDayCloses` instead would work too, but pushes an invariant into a helper that currently only filters weekends. Either way the historical rows need collapsing, and the 94 disagreeing pairs need a deliberate tie-break rather than "whichever sorts first".

10.2 Every intra-session evaluation counts the current day twice — live

Mechanism. The gatherer inserts a row for today as soon as it runs. Early in the session that row carries the previous close. It is a weekday, so `toTradingDayCloses` keeps it, and `closes` ends with the same number twice.

This is established, not inferred: §7.1 reconciles a real signal to eleven of eleven fields *only* when the duplicate is modelled, and fails on Bollinger, MACD, the probability and the forecast band when it is not.

Impact, measured on that signal: %B 0.2645 → 0.2867 (+8.4%), MACD histogram −2.1485 → −1.5954 (25.7% toward zero), composite score 76 → 75. RSI is exactly unaffected. The price bands move by less than 0.001%, because a single zero return barely moves a 252-day sample σ.

This is smaller than Finding 1 and structurally different: Finding 1 is a data defect with a clear fix, this is a semantic question about what "today's close" means before the close has happened. It also interacts with gate 4 of the DIP path, since `previousClose` is read from the same array — in the audited case the duplicate is what made `previousClose` 173.82 rather than an intraday value, which happens to be the correct comparison for an up-day test.

10.3 Suggested size crosses a currency boundary unlabelled — live

Mechanism. `getCashBalance` returns `balanceInBaseCurrency` — USD. It is passed as `budget` to `volatilityTargetedAmount`, whose result is stored raw in `SignalLog.suggestedAmount` and rendered by `formatMessage` as `Suggested: 66.85`, with no unit, immediately below `Price: 462.5 DKK`.

Extent. 85 of the 147 BUY signals in the log are quoted in a currency other than USD — EUR (61), SEK (10), NOK (5), DKK (4), CAD (2), CHF (1), HKD (1), GBp (1).

The same suggestion, four ways, from real logged signals. Converted at the rates stored in `MarketData` *on 2026-08-27.*

Symbol	Price line	Suggested line	Read natively	Intended	Error
VALE	14.955 USD	66.85	4.47 shares	4.47 shares	**1.00×**
PGHN.SW	726 CHF	66.85	0.092 shares	0.074 shares	1.24×
COLO-B.CO	462.5 DKK	66.85	0.145 shares	0.928 shares	**6.42×**
SAGA-B.ST	169.5 SEK	66.85	0.394 shares	3.760 shares	**9.53×**

The Swedish case is out by 9.5×. The single `GBp`-quoted signal is worse still: at USDGBP 0.7366 a dollar is 73.66 pence, so reading the number in the currency the line above it names understates the position by **73.7×**.

The severity is bounded by the fact that a human reads the message and places the order — this misleads rather than mis-executes. But it misleads in the one field the message exists to communicate, and the failure is silent: 66.85 is a plausible-looking number in every one of those currencies.

10.4 Logged BUY levels are not fee-adjusted; the exit machine's are — inconsistency

`toSignalLogInput` derives stopLoss $= P(1-s)$ and takeProfit $= P(1+g)$ directly from the percentages. `adaptiveTakeProfitLevel`, used by the exit machine, applies the target to $\bar{P} + f$ where f is the round-trip commission per share, so the target clears the fee as well as the profit goal.

The consequence is not cosmetic: the Simulation reads `buyRow.takeProfit` when it matches lots, so simulated exits use a target that is slightly easier to reach than the one a real active trade would face. On the audited DVA signal the difference is $f = \$2.94/1.19$ shares $= \$2.47$ per share — the fee-adjusted target would be 224.03 rather than 220.90, **1.4% further away**. Small, systematic, and in the optimistic direction.

10.5 A class-level doc comment states the opposite of the engine's direction — documentation

`leader-screen.service.ts` opens with:

> "This is the engine's PRIMARY buy path as of 2026-08-21. It exists because the composite score … rewards weakness by construction … No amount of exit tuning fixes an entry that systematically prefers falling stocks, so this screen deliberately runs the opposite sign."

That was the intention on 21 August. By 24 August the engine's own event study had shown the dip entry beating the base rate at every horizon and the leader breakout beating it at none, and the leader screen was demoted to a research surface — which is what it actually is in the code beneath the comment. The operations guide records the reversal correctly; the class comment does not, and it is the first thing a reader of that file sees.

10.6 A constant survives its purpose — tidiness

`SIGNAL_TREND_TEMPLATE_PREFERRED_RS = 90` was the leader alert's RS floor. When the alert moved to breakouts-only that floor was removed, correctly — criterion 8 already guarantees RS ≥ 70, and a breakout is self-limiting where at-pivot is not. The constant remains defined and is no longer read on that path. `SIGNAL_TT8_ALERT_MIN_RS = 90` is the live gate for the entrant alert. Two constants, the same value, one of them dead; a reader auditing the leader path will find the wrong one first.

10.7 Statistical caveats that apply to every performance number in this document

None of these is a defect. All of them bound how much any measured edge should be believed.

- **The universe is survivor-biased.** It is today's index membership plus a curated watchlist, applied to five years of history. Every absolute number — the base rate included — is optimistic; only comparisons *between* rows carry information.
- **The window is one broadly rising market** (2021-08 to 2026-08). No strategy here has been tested through a sustained bear market.
- **Exposure is not controlled for.** The headline "−175pp versus buy-and-hold" compares a strategy in the market 4–8% of the time against one in it 100% of the time. Over a window in which PLTR returned

+2,072%, that gap alone produces the entire deficit. It is an artefact of the comparison, not a measurement of pick quality.

- **Multiple comparisons are uncorrected.** The corrected Minervini audit tested 63 cells; the one that cleared $t = 2$ is roughly what chance produces. No deflation, no family-wise correction, no reality check.
- **The mid-cap universe was selected on fundamentals**, so any backtest over those names inherits the selection. Forward signals on them are clean.
- **The VCP detector is a proxy for a discretionary pattern** and will disagree with a chartist's eye in both directions.
- **Volume is measured, participation is inferred.** The engine sees how much traded, never who traded.

11 · Where to take it next

Ordered by expected value per unit of effort, with sources. The first two are corrections; the rest are extensions.

11.1 Fix the series before tuning anything else

Findings 1 and 2 sit underneath every measurement in this document. Any parameter tuned against the current series is tuned against a series that double-counts a third of its recent sessions on 527 symbols. Re-running the exit-mode comparison, the event study, or the entrant-rate calibration before de-duplicating would produce numbers that look identical in form and mean something different. This is not a judgement about which fix to choose — it is a statement about ordering.

A cheap structural guard is worth adding regardless of the fix chosen: a test asserting `count(*) = count(distinct date_trunc('day', date))` per symbol. The current unique constraint expresses the wrong invariant, and the failure was silent for four days across half the database.

11.2 Replace the flat 7.5% leader stop with an ATR-scaled one

The engine has already measured that `SIGNAL_LEADER_STOP_PCT = 0.075` is harmful in this universe: applied to the base rate it moves the 126-day median from +6.04% to −7.62% (§9.3). It also already computes a Wilder-smoothed ATR that nothing reads. The DIP path's bands are volatility-scaled; the LEADER path's are not. Making them consistent uses machinery that exists.

The practitioner convention is 1.5–2× ATR for short holds, 2–3× for swing horizons, 3–4× for trend-following — chosen so the stop sits outside normal noise while standardising dollar risk across names of different volatility. The literature here is trade press rather than peer-reviewed, so it should be treated as a starting range to be measured against this universe, not as evidence.

11.3 Wire the fundamentals gate that is already fetched

CAN SLIM's *C* and *A* — current and annual earnings acceleration — are absent from the leader screen, and `FundamentalsService` already retrieves earnings growth and ROE. This is the largest identified gap between what the method claims and what the screen implements, and it is mostly wiring.

The underlying anomaly is among the most durable in the literature. Bernard and Thomas showed that the return spread between top and bottom deciles of standardized unexpected earnings was positive in 41 of 48 quarters, with SUE-decile portfolios generating roughly 8–9% per quarter before costs — and attributed it to investors failing to recognise the implications of current earnings for future earnings. Post-earnings-announcement drift has survived four decades of scrutiny.

Note the horizon fit: PEAD plays out over one to two quarters, which matches the Trend Template's measured 126-day edge far better than it matches a 0–3 week swing. It belongs in the shortlist, not the trigger.

11.4 Reconsider the RS construction

Two changes, both cheap, both grounded.

First, the engine already computes `momentum12m2` — the 12-month return excluding the most recent month — and does not use it. The 12-2 convention is the standard one precisely because skipping the most recent month avoids short-term reversal, and skipping it enhances momentum profits. The current `rsScore` puts 40% of its weight on the trailing quarter, which loads directly on the reversal effect that convention exists to avoid. Whether IBD's responsiveness or academic momentum's cleanliness is the better fit here is an empirical question the engine is equipped to answer.

Second, the effect itself is well established — Jegadeesh and Titman's original result was that buying the top decile and selling the bottom decile of 12-month performers, held three months, returned about 1.49% per month — but three decades of follow-up have also mapped its failure modes, notably momentum crashes at market turning points. Any RS-gated entry inherits those.

11.5 Correct for multiple testing before promoting anything

The corrected Minervini audit tested 63 cells and honoured its own $n \geq 200$ pre-registration, which is already better discipline than most. The next step is to make the correction quantitative rather than narrative.

The **deflated Sharpe ratio** adjusts the significance threshold for the number of trials, the sample length, and the skewness and kurtosis of the return series — correcting for selection bias, backtest overfitting and non-normality together. For the "is this rule better than that rule" question specifically, **White's Reality Check** and **Hansen's Superior Predictive Ability** test the best rule in the context of the full universe of rules considered; Hansen's version avoids the least-favourable-configuration conservatism of White's and is more powerful in most circumstances. Both have an established track record applied to exactly this problem — technical trading rules.

11.6 Purge and embargo any future cross-validation

If parameter selection ever moves beyond hand-tuning, standard k-fold is actively misleading on this data: labels formed over a 42-day horizon overlap in time, so a training fold can contain the future of a test fold. **Purging** removes training observations whose label window overlaps the test set; **embargoing** additionally drops a fixed fraction after each test fold to handle serial correlation and market-reaction lag. **Combinatorial purged cross-validation** generates multiple train/test paths and is the standard defence against selecting a strategy on a single lucky split.

The engine already applies the same instinct in one place — `CrossSectionalService`'s `asOf` parameter and its regression test that appending future bars cannot change a past ranking. That discipline needs to extend to any parameter search.

11.7 Address survivorship properly, or bound it

Today's index membership applied to five years of history is the textbook case. The magnitude is known from CRSP: annualised returns of 7.4% survivorship-free against 9.0% survivorship-biased over 1926–2001, a 1.6pp gap — and that is for a broad market, not a screen that selects on strength.

A full point-in-time universe with delisted returns is out of reach here. The pragmatic alternative is to **bound** the bias: run the study on a known survivorship-free subset and compare against the full universe; the difference gives a rough magnitude. That is achievable with what is already stored.

11.8 Smaller items

- **Yang-Zhang for entries, not just exits.** It is 8–14× more efficient than close-to-close and, unlike Garman-Klass, unbiased in the presence of overnight gaps. The engine uses it only for owned active trades. Now that `OhlcBar` is current for 864 symbols, the reason for the asymmetry has expired — and the discontinuity when a position is flagged active is a real wart.
- **Two horizons, one message.** The levels are sized on 42 days and the probability is quoted over 252. Both are defensible; the message says "in 252 trading days" without saying the stop and target are two-month levels.
- **Sell-side rules.** Minervini's exit framework — 50-day violations, distribution days, climax action — is absent. The engine models a flat stop and a crude trail.
- **Entry at the close, not the pivot.** Buying the breakout day's *close* systematically buys extended relative to buying the pivot, and is one of the four reasons the original at-pivot study did not carry.
- **The SELL branch has never executed in production** (§6.4). Five positions have reached the trailing phase and none has exited. A dry-run harness that replays a synthetic drawdown through the live code path would be cheap insurance.

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12 · Appendices

A · Every constant that shapes a signal

Read out of the compiled `config.js` at audit time, not transcribed from source.

A.1 — Horizons and windows

Constant	Value	What it controls
SIGNAL_HORIZON_DAYS	42	the band $b = \sigma\sqrt{H}$ behind buy level, target, stop, trail
SIGNAL_FORECAST_HORIZON_DAYS	252	reach probability and the forecast band — a different horizon
SIGNAL_VOLATILITY_LOOKBACK_DAYS	252	σ is pinned to this, not to the caller's array length
SIGNAL_HISTORY_FETCH_DAYS	520	calendar days fetched; ≈ 371 weekdays, leaving margin over the 252 the longest window needs

A.2 — Entry

Constant	Value	What it controls
SIGNAL_DEFAULT_BUY_DROP_PCT	0.10	fixed dip leg, stocks
SIGNAL_BUY_SIGMA_MULT	1.5	sigma dip leg, stocks
SIGNAL_ETF_BUY_DROP_PCT	0.05	fixed dip leg, ETFs
SIGNAL_ETF_BUY_SIGMA_MULT	1.0	sigma dip leg, ETFs
SIGNAL_BUY_SCORE_MIN	55	DIP gate — the one Finding 1 perturbs
SIGNAL_NEWS_BUY_FLOOR	−0.2	news risk gate; inert while unconfigured
SIGNAL_REVERSAL_RSI_MAX	70	REVERSAL ceiling — not already extended
SIGNAL_REVERSAL_VOLUME_RATIO	1.3	capitulation volume multiple

A.3 — Exit

Constant	Value	What it controls
SIGNAL_TAKE_PROFIT_FLOOR_PCT	0.134	minimum target gain, so calm names still clear costs
SIGNAL_TAKE_PROFIT_VOL_MULT	1.5	target = 1.5 bands
SIGNAL_STOP_VOL_MULT	2	stop = 2 bands
SIGNAL_TRAIL_VOL_MULT	1	trail = 1 band below the peak
SIGNAL_EXIT_MODE	trailing	the live machine; `hold-with-stop` is the alternative
SIGNAL_HOLD_TRAIL_PCT	0.22	used only in `hold-with-stop`
SIGNAL_NOTIFICATION_COOLDOWN	6 h	per-symbol Telegram cooldown

A.4 — Sizing and cost

Constant	Value	What it controls
SIGNAL_MIN_POSITION_USD	400	just past the $380 fee-parity point
SIGNAL_BACKTEST_POSITION_SIZE	550	also the flat LEADER suggestion
SIGNAL_BACKTEST_SLIPPAGE_BPS	10	modelled slippage per leg
SIGNAL_NORDNET_COMMISSION_CLASS	MINI	0.25% + 9 SEK non-Nordic, additive
SIGNAL_SEK_PER_USD_FALLBACK	9.4645	offline paths only; live callers pass a real rate
SIGNAL_MAX_ANNUAL_VOL	0.90	strategy eligibility cap
SIGNAL_STRATEGY_SCORE_FLOOR	45	EV-ranking eligibility — lower than the DIP gate
SIGNAL_STRATEGY_MAX_FEE_RATIO	0.10	basket is trimmed until fees clear this
SIGNAL_STRATEGY_REDUNDANCY_THRESHOLD	0.40	correlation above this halves the score
SIGNAL_RECENT_SIGNAL_WINDOW	14 d	recently-exited exclusion

Constant	Value	What it controls
SIGNAL_TREND_TEMPLATE_MIN_RS	70	criterion 8 — the definition of 8/8
SIGNAL_TREND_TEMPLATE_PREFERRED_RS	90	**defined, no longer read on the leader path** (§10.6)
SIGNAL_TREND_TEMPLATE_MIN_PASSES	8	all eight
SIGNAL_TREND_TEMPLATE_SMA200_RISING_DAYS	21	a month of non-decreasing 200-day
SIGNAL_TREND_TEMPLATE_MIN_ABOVE_LOW_PCT	0.30	criterion 6
SIGNAL_TREND_TEMPLATE_MAX_BELOW_HIGH_PCT	0.25	criterion 7
SIGNAL_VCP_SWING_THRESHOLD_PCT	0.02	ZigZag sensitivity
SIGNAL_VCP_MIN/MAX_CONTRACTIONS	3 / 6	base shape
SIGNAL_VCP_MAX_FINAL_TIGHTNESS_PCT	0.10	measured median is 3.5%, so this is loose by design
SIGNAL_VCP_MIN/MAX_BASE_DAYS	20 / 60	base length
SIGNAL_VCP_MAX_BASE_DEPTH_PCT	0.35	added 2026-08-24; previously unbounded
SIGNAL_VCP_MAX_DRYUP_RATIO	0.85	**hard gate** — above it there is no VCP
SIGNAL_VCP_BREAKOUT_VOLUME_RATIO	1.40	Minervini's "40–50% above average"
SIGNAL_VCP_PIVOT_PROXIMITY_PCT	0.02	one-sided since 2026-08-24
SIGNAL_VCP_MERGE_RALLY_PCT	0.50	merge threshold — the core of the rewrite
SIGNAL_LEADER_STOP_PCT	0.075	flat stop — measured harmful in this universe (§9.3)
SIGNAL_LEADER_ALERT_COOLDOWN_DAYS	5	a breakout bar keeps testing true for days
SIGNAL_TT8_ALERT_MIN_RS	90	13.4 entrants/day → 3.9
SIGNAL_TT8_COOLDOWN_DAYS	10	stops boundary-hovering names re-alerting
SIGNAL_SHORTLIST_MAX	12	digest cap, overflow summarised
SIGNAL_MARKET_BREADTH_HEALTHY_PCT	0.50	the natural line, deliberately not fitted
SIGNAL_SCREEN_MIN_DOLLAR_VOLUME	$5M	50-day turnover floor for leader signals
SIGNAL_RS_WEIGHTS	0.4 / 0.2 / 0.2 / 0.2	3m double-weighted, IBD convention
SIGNAL_RS_MIN_UNIVERSE	30	below this, no percentile is claimed

B · File map

Concern	File	Lines
Orchestrator, all delivery, all formatting	`signals/signals.service.ts`	5,399
Tier 1 indicators, score, adaptive levels	`signals/indicators.service.ts`	684
Tier 2 forecast, probability, sizing	`signals/forecast.service.ts`	239
Trend Template, VCP, alert selection	`signals/leader-screen.service.ts`	691
Cross-sectional RS, peer ranks	`signals/cross-sectional.service.ts`	312
Market direction	`signals/market-breadth.service.ts`	154
Bar store, refresh-range logic	`signals/ohlc-bar.service.ts`	269
Frozen-target tracking for real fills	`signals/signal-trade-tracking.service.ts`	556
EV ranking, basket sizing	`signals/strategies.service.ts`	726
Backtest harness	`signals/backtest.service.ts`	459
Simulation curves	`signals/simulation-performance.ts`	244
Commission — single source of truth	`libs/common/src/lib/nordnet-fees.ts`	229
Every threshold	`libs/common/src/lib/config.ts`	—
Weekday filter, series metrics	`signals/fund-history.service.ts`	1,032
Schedule	`services/cron/cron.service.ts`	—

C · Reproducing this audit

```
# 1 · compile the services the scripts load
npx tsc -p tsconfig.signals-test.json

# 2 · pull the exemplars and re-run the engine's own maths against them
node extract-evidence.cjs evidence.json

# 3 · quantify Finding 1
node measure-duplicates.cjs duplicates.json

# 4 · typeset
node build-audit-pdf.cjs audit.html ENGINE_AUDIT_2026-08-27.pdf

# supporting checks used in §10
node run-signals-tests.cjs              # 313 unit tests
node run-explain-symbol.cjs DVA PGHN.SW # per-symbol screen breakdown
```

All three audit scripts open a read-only connection. Nothing in this audit wrote to the database, and no repository file was modified in producing it.

D · What would change this report

Stated so the report can be falsified rather than merely believed.

- Finding 1 is wrong if `MarketData`'s duplicate rows are somehow collapsed before reaching `toTradingDayCloses`. They are not — `measure-duplicates.cjs` reads through the same helper the engine uses, and gets a different array length from the deduplicated one for 397 symbols.
- Finding 2 is wrong if the DVA reconciliation could be achieved another way. Six alternatives were tried — exclusive cut-off, inclusive cut-off, live price appended, raw calendar series, full history, and today's table —

and none reproduces all four snapshot fields. Appending one duplicate of the final close reproduces all eleven.

- Finding 3 is wrong if `getCashBalance` returns a native-currency figure. It returns `cashDetails.balanceInBaseCurrency`.
- Every performance table in §9 is quoted from the engine's own study scripts and inherits their caveats (§10.7). None was re-run for this audit, and re-running them against a de-duplicated series is exactly what §11.1 asks for.